

Revisiting the Effects of Health on Retirement Expectations*

Aspen Gorry[†] Jonathan M. Leganza[‡]

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Abstract

How does health affect retirement expectations? We revisit this question using updated data from the Health and Retirement Study and an event study framework to estimate the causal effects of various health shocks on the self-assessed probabilities of working past older ages. Importantly, recent waves of survey data contain information on expectations for current workers and non-workers, which avoids sample selection issues encountered by previous studies. We find that declines in subjective health status and new diagnoses of objective health conditions decrease the probabilities of working past 62 and 65. Our findings highlight how poor health can cause people to expect to retire before becoming eligible for Social Security and Medicare benefits.

Keywords: retirement expectations, retirement timing, health shocks

JEL codes: J26, J22, I10

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[†]John E. Walker Department of Economics. Clemson University. (email: aspen.gorry@gmail.com)

[‡]John E. Walker Department of Economics. Clemson University. (email: jleganz@clemson.edu)

1 Introduction

The timing of retirement is a major determinant of lifetime earnings and a crucial factor in financial security at older ages. However, people face uncertainty about the timing of their retirement and make consumption and savings decisions based on their expectations. One prominent source of uncertainty relates to the evolution of health. While declines in health may decrease work capacity and prevent people from working as long as they had planned, some health conditions may lead to greater medical expenses, generating an incentive to work longer (especially when health insurance is tied to employment). Of course, the relative strength of these competing forces is likely to depend on the specific nature of the health condition. Therefore, an important task for empirical work is to provide evidence on how changes in health influence retirement expectations and to document how the magnitudes of the effects vary across different types of health risks.

Yet, providing causal evidence on how health affects retirement expectations is difficult. There are two key challenges. One is identification. Correlations between health and expectations may not reflect causal relationships because of concerns like omitted variable bias. For example, unobserved preferences for leisure can influence how long people plan to work, but also health behaviors and investments. The other challenge is data availability. The Health and Retirement Study has served as a dominant source of data on retirement expectations for decades, but early waves of the survey collected information on these expectations only for people currently working, which is a selected sample. This data limitation prevented researchers from analyzing expectations for an important group, those whose health prevented them from working when they were surveyed.

In this paper, we use an event study framework and updated data to overcome these challenges, providing new causal evidence on how health impacts retirement expectations. For identification, we leverage the quasi-random timing of various health shocks. For each health shock of interest, we focus on a group of people who all experience the same shock and then track the evolution of retirement expectations around the timing of the shock. Our approach thus avoids potentially problematic comparisons between people who experience declines in health and people who do not. Moreover, recent waves of the Health and Retirement Study ask both current workers and non-workers about their retirement expectations. We avoid sample selection concerns by conducting our event study analysis using these updated data.

We study several different analysis samples, one for each health shock of interest. Our data allow us to study shocks defined using both subjective and objective health measures.

For each shock, we define a sample of people between the ages of 50 and 60 who experience the shock and analyze how their self-assessed probabilities of working past older ages evolve before and after the shock. Our primary outcome variables are the self-assessed probabilities of working past 62 and 65, but we also study the probability of working past 70 when possible. The first two outcomes are especially relevant for policy because they correspond to eligibility ages for Social Security retirement benefits and Medicare, respectively.

We begin by studying declines in subjective health. Specifically, we estimate the effects of a decline in self-reported health status from excellent, very good, or good to fair or poor. The event study estimates reveal flat pre-shock trends in expectations before the health status decline and large decreases in the likelihood of working past 62 and 65 after the shock. Our preferred specification indicates that a decline in health status reduces the likelihood of (i) working past 62 by 4.1 percentage points, an 8.9% decrease from the baseline mean of 46%, and (ii) working past 65 by 4.5 percentage points, a 13.6% decrease.

Next, we study declines in objective health, focusing on two main measures. First, we estimate the effects of overnight hospitalizations. The signs of the estimates suggest that hospitalizations lead people to expect to retire earlier, but the estimates are not statistically different from zero. Second, we estimate the effects of a newly diagnosed health condition. In the spirit of Hosseini, Kopecky and Zhao (2022), we construct a summary measure equal to the total number of conditions a person has ever experienced, using variables for diagnoses of arthritis, cancer, diabetes, heart attacks or heart disease, high blood pressure, lung disease, psychiatric conditions, and strokes. We then define a new diagnosis shock as an increase in this summary measure. We find that new diagnoses lead to statistically significant decreases in the likelihood of working past 62 and 65 that amount to 2.8 and 1.7 percentage points, respectively, which are decreases of 6.7% and 5.9% when compared to the means.

We then unpack this result on new diagnoses by studying separate events for each of the eight health conditions in our summary measure. Some, but not all, of these conditions significantly impact retirement expectations. For example, we find clear evidence of decreases in the likelihood of working past 65 after diagnoses of cancer and strokes. The point estimates for these shocks indicate sizable reductions in the probability of working of 20.0% and 44.6% when compared to the means. In contrast, we find little to no evidence that new diagnoses of diabetes and high blood pressure lead to changes in retirement expectations.

Our study relates most closely to papers that analyze health and retirement expectations (Dwyer and Mitchell, 1999; McGarry, 2004; Gupta and Larsen, 2010; Munnell, Sanzenbacher and Rutledge, 2018; Caliendo et al., 2023; Giustinelli and Shapiro, 2024). Two of these pa-

pers, Dwyer and Mitchell (1999) and McGarry (2004), stand out as the most similar to ours.¹ They use the initial HRS survey waves and regression analyses to establish important links between several health measures and retirement expectations for workers. But causal interpretations of the estimates in these earlier studies could be threatened by sample selection because the expectations variables only exist for workers in the earlier survey waves. McGarry (2004) summarizes the sample selection problem by noting that the “drawback of this methodology is that because the expected probability of full-time work is available only for those still in the labor force, the sample is a selected one.”

Our main contribution is to use a quasi-experimental framework and updated data to produce new causal evidence on how health shocks affect retirement expectations. Two features of our study allow us to advance the literature in this way. First, we avoid sample selection problems by using only the more recent waves of HRS data that include well-defined outcomes, even for nonworkers. Specifically, beginning with wave 8, the survey consistently asks everyone—not just workers—about the likelihood that they work past 62 and 65. The newer data thus allow us to track changes in expectations for a crucial group of people absent from data covering only workers: those who (temporarily or permanently) stop working because of their health. If health shocks cause some people to (i) stop working immediately and (ii) believe they are less likely to work past 62 and 65, then non-workers dropping out of the data after health shocks would bias estimates upwards. Consistent with this idea, when we conduct our analysis on a selected sample of workers, we find no statistically significant evidence that health shocks cause workers to expect to retire earlier. Second, our event study framework allows us to leverage the quasi-random timing of shocks, cleanly isolate the effects of different shocks, and produce graphical evidence that allows for transparent assessments of the key identification assumptions.

In addition to these methodological advancements, we provide estimates for additional outcomes (the probabilities of working past 65 and 70) and for birth cohorts approaching retirement during the more recent period, which is important because the effects of health on retirement expectations are likely to depend on setting. For instance, increasing capacity to

¹The other papers take different approaches. Gupta and Larsen (2010) use data from Denmark to study how the relationship between health and retirement expectations varies when using administrative versus survey data. Munnell, Sanzenbacher and Rutledge (2018) study how various factors, including health, contribute to earlier-than-planned retirements, and Caliendo et al. (2023) study how health and other factors influence the difference between expected and actual retirement ages. Finally, Giustinelli and Shapiro (2024) estimate person-specific “subjective ex ante treatment effects” to uncover the effects of health on working by comparing a person’s own estimate of the probability they work in poor health to their estimate of the probability they work in good health.

work at older ages, greater opportunities to work from home, and changes to the retirement policy landscape (e.g., the decline of defined benefit pensions) may influence how people respond to health shocks.

Our paper also naturally connects to a much broader literature on the effects of health on retirement and labor supply at older ages (e.g., McClellan, 1998; Bound et al., 1999; Blau and Gilleskie, 2001; Coile, 2004; Disney, Emmerson and Wakefield, 2006; García-Gómez et al., 2013; Seligman, 2014; Gustman and Steinmeier, 2018; Blundell et al., 2023; Hsu, Morrill and Pathak, 2024). Several papers provide reviews on this longstanding question (e.g., Currie and Madrian, 1999; Coile, 2016; O'Donnell, Van Doorslaer and Van Ourti, 2015; Blundell, French and Tetlow, 2016; French and Jones, 2017).² While the methods and data vary across studies, most of the related reduced form papers show that declining health leads to declines in employment.

To the extent that one can interpret changes in expectations after health events as changes in the timing of future retirements, our analysis suggests that people who experience health shocks in their 50s ultimately retire earlier than they otherwise would have. Importantly, earlier evaluations of the HRS expectations data indicate that, on average, people have rational expectations about retirement timing (Benitez-Silva and Dwyer, 2005) and that expectations strongly predict future retirements (Haider and Stephens Jr., 2007). Moreover, recent evidence confirms these ideas and highlights how the subjective probabilities of working past older ages are especially useful unbiased predictors of actual retirement behavior on average (Kézdi and Shapiro, 2023).

With this interpretation, our findings on specific diagnoses connect to research on health and income inequality. O'Donnell, Van Doorslaer and Van Ourti (2015) highlight that poor health influences individual income through employment and Hosseini, Kopecky and Zhao (2024) show that differences in health measured using a frailty index account for about a quarter of the variation in lifetime earnings. Our results highlight which conditions appear to induce earlier retirements, demonstrating which conditions may be the most important contributors to shorter careers and earnings inequality. Still, our outcome variables are not realized retirements, so we caution against strict interpretations of our results as changes in retirement.

²A separate but related literature studies the relationship between health and retirement in the opposite direction and asks how retirement impacts health (e.g., Coe and Zamarro, 2011; Coe et al., 2012; Eibich, 2015; Gorry, Gorry and Slavov, 2018; Nielsen, 2019; Gorry and Slavov, 2021, 2023).

2 Data

We use data from the Health and Retirement Study (HRS). The HRS is a longitudinal, biennial survey covering Americans over 50 and their spouses. It consists of seven sample cohorts based on the date of their first interview. The first of these “HRS cohorts” was initially interviewed in 1992; the most recent cohort was initially interviewed in 2016. To access the data, we use the RAND HRS Longitudinal File 2022 (v1) dataset (Bugliari et al., 2025), which is a cleaned and streamlined HRS product from the RAND Center for the Study of Aging that includes key information on all survey cohorts and every person interviewed.

These data are well-suited for our analysis for two reasons. First, the breadth of the survey allows for a thorough analysis. Crucially, the data contain information on retirement expectations and also detailed information on health. Second, the survey’s focus on older people produces a sizable sample of individuals approaching retirement age.

2.1 Outcome Variables: Retirement Expectations

The outcomes in our analysis capture expectations about retirement timing. Specifically, we study self-assessed probabilities of working past older ages. The HRS contains variables for the probabilities of working full-time after 62, 65, and 70. The values are recorded on a scale from 0 (“absolutely no chance” of working past that age) to 100 (“absolutely certain” about working past that age) and the values for working past 65 and 70 are set to 0 if the person said that there was no chance they were going to work full-time past one of the earlier ages. We divide the values by 100 to reflect probabilities.

We focus mostly on the probabilities of working past 62 and 65 because these variables are available in all survey waves.³ They are also especially relevant because they correspond to eligibility ages for important government programs; age 62 is when people become eligible to claim old-age benefits from Social Security, and age 65 is when people become eligible for health insurance through Medicare. In contrast, while we also study the probability of working past 70, this variable is available only from wave 11 onward, which limits its use.

Figure 1 presents histograms of these outcome variables. The underlying data contain people between ages 50 and 60 in the later survey waves (8 through 16), who form the basis of our analysis samples. For the probabilities of working past 62 and 65, the most common responses are 0, 50, and 100 percent, although there is mass throughout the distribution.

³In general, these reported probabilities appear internally consistent. For example, more than 99% of the individuals in our analysis data always report a probability of working past 62 that is greater than their probability of working past 65.

The graph for the probability of working past 70 looks different (and is on a different scale). About 60% of people report no chance of working full-time past 70, and only 1% report that they will work past this advanced age with certainty.

An advantage of these probability-based outcomes is that they capture different types of changes in retirement expectations. For example, consider a person who originally plans to retire at 65 and who then experiences a health shock. One possibility is that this person updates their plans to retire at a different age. Another possibility is that this person continues to plan to retire at 65, but they may be less confident in their ability to do so. Our outcomes should capture each of these important types of changes.

The main disadvantage of these variables used to be that there were many missing values. The underlying survey questions were only asked to workers in earlier waves. However, starting with wave 8 (corresponding to 2006), the questions were asked regardless of work status. Appendix Figure A.1 illustrates this point by plotting the fraction of observations missing values for each of the outcomes by work status across waves. For the two main outcomes, values are mechanically missing for non-workers in waves 1 and 3 through 7. From wave 8 onward, the rates of missing values are similar for workers and non-workers.⁴

However, in wave 15, which corresponds to 2020, there is a notable increase in missing values for workers, most prominently for the probabilities of working past 65 and 70. This increase is due to a skip pattern error in the survey that meant that respondents interviewed in early months (especially March, April, and May) erroneously skipped the questions if they were between 62 and 64 or if they were currently working. While these missing values could potentially be problematic because it introduces a sample selection concern, few people in our shock-specific analysis samples, who, importantly, are all 60 or younger, were interviewed during these months. Nonetheless, we conduct a robustness check later that excludes all observations of individuals interviewed during these months and find similar results.

There is still other missing data. For example, consider the probability of working past 62. About 5% of these values are missing for people between 50 and 60 in the later survey waves. Some values are missing due to proxy interviews (about 2.5% of observations). For survey participants unable or unwilling to do an interview, the HRS offers the opportunity to use a proxy respondent (usually a family member like a spouse or adult child). In these cases, the expectations section of the questionnaire is skipped. Other values are missing because the questions are skipped for people who had difficulty with other probability questions asked

⁴The data contain another retirement expectations variable: the expected retirement age. While it would be interesting to analyze too, it is still mechanically missing for non-workers.

earlier in the survey (0.6% of observations), because the respondent did not know the answer to the question (1%), or because of some “other” reason (1%).

In our baseline analysis, we use all available person-wave observations with non-missing values, but we check the sensitivity of our estimates to missing data in the robustness section in three ways. First, we limit our analysis to only people who have no missing values for the outcomes. Second, we limit our analysis to a balanced panel. Third, we set the probabilities of working past 62, 65, and 70 to zero when they are missing because of a proxy interview. The idea is to address the largest source of missing data and to account for the fact that these missing expectations are for people who experience a health shock that is severe enough to lead to a proxy interview, making continued work less likely.

2.2 Health Variables Used to Define Health Shocks

We use several health variables to define the health shocks we study and their corresponding analysis samples, which we detail in the next subsection. To study subjective health, we use a categorical variable that captures whether the individual considers their health to be poor, fair, good, very good, or excellent. This variable has known limitations. It measures health using an ordinal, but otherwise qualitative, scale, which can limit comparisons across people or even within people over time (if, for example, people judge their health status differently when they are younger versus older). We acknowledge that this health measure is far from perfect. Still, it is commonly used in the broader literature, and it has the advantage of being able to detect meaningful changes in health that are not accompanied by objective diagnoses or other verifiable information.

To study objective health, we use one indicator variable that captures whether the individual has been hospitalized overnight since their last interview, and another set of indicator variables that capture whether a doctor has ever told the individual that they have a specified health condition. We use variables for diagnoses of (i) arthritis, (ii) cancer, (iii) diabetes, (iv) heart attacks or heart disease, (v) high blood pressure, (vi) lung disease, (vii) strokes, and (viii) psychiatric problems.⁵ One advantage of these objective measures is that they should reflect verifiable conditions and are thus not limited by the same set of concerns as

⁵The survey question for heart disease refers to heart attacks, coronary heart disease, angina, congestive heart failure, or other heart problems. The question for cancer refers to cancer or a malignant tumor of any kind except minor skin cancer. The question for lung disease refers to chronic lung disease such as chronic bronchitis or emphysema, but not asthma. The question for arthritis refers to arthritis or rheumatism. The question for diabetes refers to diabetes or high blood sugar. The question for high blood pressure refers to high blood pressure or hypertension. The question for psychiatric problems also refers to any emotional or nervous problems.

the subjective measure of health status. Of course, we are using survey data, so the objective conditions that we study are still self-reported. Receiving an official diagnosis also requires seeking medical care, which means that these measures are not well-suited for detecting changes in health for people who choose not to undergo medical tests.

In our main analysis, we combine these diagnosis indicators into an index by summing them to create one measure that captures the total number of diagnosed health conditions that a person has. We use this measure, which is similar to frailty indices used in gerontology (e.g., Searle et al., 2008) and recently in economics (Hosseini, Kopecky and Zhao, 2022), to study a general shock defined as a new diagnosis. Using a summary measure like this has advantages. It is a useful way to aggregate information, allowing us to pool data on individuals who all experience a new diagnosis. The tradeoff is that by including all conditions in the index, the sample of people who experience a new diagnosis can develop different health conditions that could generate different responses. Therefore, we also analyze each condition-specific diagnosis separately.

2.3 Constructing Analysis Samples

We construct several analysis samples based on the shocks that we study. We begin with the RAND Longitudinal File, which includes everyone interviewed in the HRS, and implement three basic sample restrictions. First, we keep only people who are alive and who responded to the survey. Second, we keep people between ages 50 and 60, who are approaching typical retirement ages and have not yet reached the specific ages referenced for the probability-based retirement expectations outcomes. Third, we only keep observations from survey waves 8 through 16 (2006 through 2022). In these waves, the retirement expectations questions are asked of all people, not just workers, which allows us to avoid the concern about sample selection.

Next, we define the analysis sample for each health event we consider. We study three main health shocks: (i) declines in health status, (ii) new diagnoses of health conditions, and (iii) hospitalizations. The health status events reflect changes in a subjective measure of health, and we define a health status event to capture new and meaningful declines in overall health. Specifically, we define a person as experiencing a health status event in survey wave w if they report their health status as either fair or poor in wave w and as either excellent, very good, or good in the two prior consecutive waves, $w - 1$ and $w - 2$.

Our main definition of this health shock is consistent with prior work and is intended to (i) capture significant changes in health and (ii) lessen measurement concerns with how people

may use the qualitative health scale differently. However, this definition might miss less severe shocks that are still relevant for retirement expectations. We therefore also analyze alternative health status events in the appendix that are designed to capture any downgrade in health status. We define a person as experiencing one of these alternative events if they have lower health status in wave w compared to the prior wave, and to be consistent, we require that they were observed two waves prior as well.

In contrast, diagnoses of health conditions and hospitalizations reflect changes in objective health measures. We define a person as experiencing a new diagnosis event in wave w if they report more diagnosed health conditions in wave w than in wave $w - 1$ and $w - 2$. Finally, we define a person as experiencing a hospitalization event in wave w if they report in wave w that they were hospitalized overnight since their last interview and report no such hospitalizations in waves $w - 1$ and $w - 2$.

These events constitute our main health shocks of interest, but the detailed nature of our data also allows us to study separate events for each of the eight underlying diagnosis variables. For each diagnosis, we define a person as experiencing the event in wave w if they report having the condition in wave w and not in waves $w - 1$ and $w - 2$.

The analysis samples for each event consist of only people experiencing the shock. If a person experiences the shock more than once, we study the first instance. To focus on the evolution of outcomes around the timing of the shock, we limit the analysis samples to observations of people up to three waves before and up to one wave after the event.

Our approach to structuring the data means that our analysis samples contain people who experience health shocks in different survey waves. Appendix Figure A.2 illustrates how many people in each of our main analysis samples experience the corresponding health shock in each survey wave. Because our data consist of waves 8 through 16 and our event definitions require that we observe the person in two pre-shock periods, wave 10 is the earliest event wave. We do not require that we observe people in waves after the shock, so we are able to include people who experience the shocks in wave 16. Note that the frequencies should in general decrease over time as individuals age out of potential inclusion in the sample, which is restricted to people between 50 and 60; however, the HRS added new survey cohorts in waves 10 and 13, which leads to increases in frequencies in waves 12 and 15, respectively, as these are the first waves for which individuals in these newly-added cohorts can experience a shock and be included in our sample.

Table 1 presents summary statistics for our three main analysis samples. For each analysis sample, the statistics in the table correspond to observations in the survey wave before the

shock. On average, people report roughly a 45% chance of working past 62, a 30% chance of working past 65, and a 12% chance of working past 70. The observation counts reflect some missing data for the outcomes. Between about 3.5% and 6.5% of values are missing for the probabilities of working past 62 and 65, whereas the fraction missing is greater for the probability of working past 70 because the underlying survey question is not asked until wave 11. The table also highlights differences in sample sizes. We have roughly three times as many observations on new diagnosis events compared to the events for declines in subjective health status. Studying separate samples for each health event allows us to cleanly isolate the effects of different shocks, but differences in our estimates (and their precision) across shocks can reflect differences in the health conditions, but also differences in the composition and size of the analysis samples.

3 Identification Strategy

We use an event study framework to analyze the effects of health shocks on retirement expectations. For each shock of interest, we exploit the timing of the shock by limiting our attention to people who experience the shock and comparing the evolution of retirement expectations before and after the shock occurs. One advantage of this approach is that we do not rely on comparisons of people who experience the shocks of interest to those who do not. By focusing on groups in which everyone experiences the same shock, we avoid identification concerns centered on the idea that people who experience a given shock may differ in unobservable ways from people who do not experience the shock. We take this “timing-based” event study approach (Miller, 2023) because we prefer the assumption that the timing of health shocks is as good as random for those who experience them over the assumption that a control group of people who do not experience health shocks provide the correct counterfactual for over-time changes in retirement expectations.

Our focus on only people who experience health shocks also means that our estimates should be interpreted as treatment effects for this group. Our estimates may not apply to the broader population. It could be that people who experience the health shocks that we study are different from the population in observable and unobservable ways. However, we argue that the group of people who do experience these shocks are a key population of interest, as these may be people who face more uncertainty about the evolution of their health.

To implement our event study analysis, we use two regression models that follow Dobkin et al. (2018), who use administrative and HRS data to study the effects of hospitalizations on

financial outcomes. We begin by using a nonparametric event study framework to graphically analyze the evolution of outcomes around the timing of each shock. For each analysis sample, we estimate equations of the following form:

$$y_{ict} = \alpha + \sum_{\tau=-3}^{-2} \delta_{\tau} + \sum_{\tau=0}^1 \delta_{\tau} + \lambda_{ct} + \varepsilon_{ict}, \quad (1)$$

where y_{ict} is an outcome variable (such as the probability of working past 65) for individual i in HRS cohort c during survey wave t , λ_{ct} is an HRS-cohort-by-wave fixed effect, δ_{τ} is a coefficient on an event time indicator for a survey wave relative to the wave during which the shock occurs. As in Dobkin et al. (2018), we include HRS-cohort-by-wave fixed effects instead of simply survey wave fixed effects to account for changes in the composition of the HRS sample as cohorts are added to the survey. Our event window includes three survey waves before the shock, the wave of the shock, and one wave after the shock. The δ_{τ} s are the coefficients of interest and capture the average difference in the outcome at event time τ relative to the omitted period, $\tau = -1$, the survey wave before the event occurs.

Note that the regression model is not a two-way fixed effects model because we do not include individual-specific fixed effects in our baseline specification. Borusyak, Jaravel and Spiess (2024) point out identification issues with two-way fixed effects event study models where all units are treated at different times. The basic idea is that the full set of event time indicators is collinear with unit and time fixed effects. Their earlier working paper (Borusyak and Jaravel, 2018) highlights how dropping the unit fixed effects is an immediate and easy-to-implement solution. We take this very simple approach in our baseline analysis.

The identification assumption underlying regression model (1) is that conditional on experiencing a given shock, the timing of the shock is uncorrelated with the outcomes. For each shock that we study, the threat to identification is thus that there is some unobserved factor that influences both the timing of the shock and retirement expectations. One concern might relate to changes in job characteristics or employment opportunities. For instance, a threat to our design is if increases in physical labor at work or increases in stress about future employment (i) cause people to update their retirement expectations but also (ii) cause people to be more likely to experience a health shock.

The ex-ante plausibility of the identification assumption may vary across shocks. For example, some health conditions may be sudden and unpredictable, whereas others may be expected based on previous conditions or gradually deteriorating health. In general, we can provide an assessment of the validity of the identifying assumption for each shock by

analyzing the estimated δ_τ s for $\tau < 0$. The patterns of these pre-period estimates provide evidence on whether outcomes were trending before the shock of interest.

We often find little to no evidence of problematic pre-trends in outcomes, but there is some evidence of pre-period trends in some cases. Therefore, to consistently account for the possibility of underlying trends when analyzing the effects of different shocks, we move away from the nonparametric regression model and instead use a parametric specification.

Specifically, we estimate

$$y_{ict} = \alpha + \beta\tau + \sum_{\tau=0}^1 \delta_\tau + \lambda_{ct} + \varepsilon_{ict}. \quad (2)$$

The key difference here is the inclusion of τ , a linear time trend. The coefficient β corresponds to the pre-shock linear trend in the outcome variable. The parameters of interest are the δ_τ s, which now capture the average difference in the outcome at event time τ compared to its linear pre-trend. The identification assumption underlying regression model (2) is that, conditional on experiencing a given shock, the timing of the shock is uncorrelated with deviations in the outcome variable from its pre-period linear trend.

When estimating the parametric event study regressions, we focus on δ_0 , the effect of the shock on the outcome in the survey wave the shock is reported. We focus on this point estimate instead of δ_1 , which corresponds to the subsequent wave, because our definition of each shock requires that we observe the individual in the data during the wave of the shock (and the two previous waves) but not afterward.

We rely on this baseline parametric specification to quantify magnitudes and assess the statistical significance of the results. Later, we assess the robustness of the estimates to changes in the regression specification and the composition of the analysis samples. Specifically, we report estimates from alternative specifications where we (i) include demographic control variables, (ii) include wave fixed effects instead of cohort-by-wave fixed effects, (iii) include individual fixed effects, (iv) use survey weights, (v) use survey weights and individual fixed effects, (vi) use only the subsample of people who are never missing values for the outcomes, (vii) use only the subsample of observations that form a balanced panel, (viii) impute values for proxy interviews, (ix) exclude observations from early 2020 survey interviews, and (x) exclude observations from waves 15 and 16, which could in general be influenced by Covid. We also assess the robustness of our estimates to using recently-developed two-way fixed effects specifications that address issues with those models when treatment effects are heterogeneous (Sun and Abraham, 2021).

4 The Effects of Health Shocks on Retirement Expectations

In this section, we present our results. For each health shock, the sign and magnitude of the effects are theoretically ambiguous. Health shocks might lead workers to update their expectations towards retiring earlier if they lead to declines in work capacity or longevity. Of course, the amount of updating in this direction is likely to depend on the extent to which the specific health shock is debilitating. On the other hand, it is at least theoretically possible that some health shocks lead workers to update their expectations towards retiring later, perhaps if they lead to increased medical expenses or temporary earnings declines. These sources of ambiguity are important to note when looking at retirement expectations rather than contemporaneous labor supply responses because some health shocks could induce immediate exits from the labor market, but people then might expect to work longer once they return to work.

4.1 The Effects of Health Status Declines, New Diagnoses, and Hospitalizations

We begin with an analysis of our three main health shocks: (i) a decline in subjective health status, (ii) a new diagnosis of a health condition, and (iii) a hospitalization. Figure 2 displays the nonparametric event study results. Each graph corresponds to one of the shocks and one of the main outcomes, either the probability of working past 62 or the probability of working past 65. The point estimates in the graphs are the δ_τ coefficients from estimating equation (1).

First, consider panels (a) and (b), which present the results for a decline in subjective health status. The pre-shock period point estimates indicate a lack of significant trends in the probabilities of working past key ages before a decline in health status, consistent with the interpretation that the underlying causes of the declines in health may occur suddenly. The post-shock period estimates show a sharp and clear decline in the average probability of working past 62 and 65. These two panels indicate that declines in subjective health status cause workers to update their expectations towards retiring earlier.

Next, consider panels (c) and (d), which present the results for a new diagnosis of a health condition. Here, we also find pre-shock period point estimates that are not statistically distinguishable from zero, although there is more of a visible downward trend before the shock. Still, the post-shock period estimates reveal a sharp and greater decline in retirement expectations after people are diagnosed with a new health condition. Finally, consider panels (e) and (f), which present the results for a hospitalization. There is some visual evidence of

a decline in the likelihood of working past 65, but not past 62.

To quantify these findings, Table 2 presents the results from the parametric event study analysis. The parametric approach summarizes the effects by comparing the evolution of the outcome variables to their pre-period trends (which were relatively flat for declines in health status and hospitalizations but were more noticeable for new diagnoses). Panel A of the table presents results for each of the main shocks. The first four columns present results for the probability of working past 62. Column (1) displays the point estimate for δ_0 from estimating equation (2) for this outcome, and the subsequent columns display the dependent variable mean in the survey wave preceding the shock, the number of clusters (unique individuals), and the number of observations in the analysis sample. The next four columns present the corresponding results for the probability of working past 65.

On average, a decline in health status causes a statistically significant 4.1 percentage point decline in the self-assessed probability of working past 62 and a statistically significant 4.5 percentage point decline in the probability of working past 65. These declines translate to meaningful reductions in the likelihood of working by 8.9% and 13.6%, respectively, compared to the baseline means. On the one hand, the pattern of the estimates, which suggests larger declines in the probability of working past 65 compared to 62, is consistent with the idea that people believe they are increasingly less likely to work as they age because of the health shock. On the other hand, the estimates are not statistically different from one another in absolute, percentage point terms (p-value = 0.781) or when we rescale the outcomes by their pre-period means and re-estimate our regressions so that the estimates reflect percent changes (p-value = 0.224).

Recall that these health status events are defined as a person declining from either excellent, very good, or good health to fair or poor health. Appendix Table A.1 and Appendix Figure A.3 present results for the alternative definition defined as any downgrade in health status, which is intended to capture less severe but potentially relevant declines in health. Consistent with this idea, the point estimates for the probabilities of working past 62 and 65 are economically and statistically significant, but smaller than the above estimates, both in absolute magnitude and when compared to the baseline means. These results support the idea that more substantial shocks translate to greater changes in retirement expectations.

New diagnoses of health conditions also lead to statistically significant declines in the probabilities of working past older ages. The point estimates in Table 2 indicate a 2.8 percentage point decline in the likelihood of working past 62, which is a 6.7% decline compared to the baseline mean, and a 1.7 percentage point decline in the likelihood of working past

65, which is a 5.9% decline compared to the baseline mean. These results are also not statistically different from one another; we cannot reject the null hypothesis that the baseline point estimates in percentage point terms are the same ($p\text{-value} = 0.195$), nor can we reject that the estimates in percent terms are the same ($p\text{-value} = 0.761$).

In contrast, we find no evidence that hospitalizations lead to changes in the probabilities of working past 62 or 65. The estimates are smaller in magnitude than those for the other health shocks, and they are not statistically significant. It could be that this lack of evidence is explained by the fact that hospitalization events may capture conditions, injuries, or medical procedures that are less debilitating or that are not expected to have long-lasting impacts on work capacity. We also find little to no evidence that any of the main health shocks impact the likelihood of working past 70 (see Appendix Figure A.4 and Appendix Table A.2), which could be due to the lower likelihoods of working past 70 to begin with; the means for this outcome range from 11% to 13%. However, recall that our analysis of working past 70 is also more limited due to smaller sample sizes because it is only available for waves 11 onward.⁶

To help contextualize these results on expectations, we compare them to changes in current labor supply. Appendix Figure A.5 and Appendix Table A.3 present the results from estimating our event study regressions using an indicator variable that takes the value of one if the respondent is currently working full-time as the outcome. The results for health status declines and new diagnoses indicate that current labor supply and expectations about future labor supply move together. For health status declines, we find strong, clear evidence of an 8.1 percentage point decline in labor supply. For new diagnoses, the graph is less clear because of pre-trends, but the parametric estimate indicates a 3.0 percentage point decline. However, the results for hospitalizations show current labor supply responses that differ from expectations. We find a large, 5.3 percentage point decline in current labor supply, which is generally consistent with the findings in Dobkin et al. (2018). However, the lack of a strong effect on expectations after hospitalizations in our main results suggests that people may not expect these contemporaneous declines to be so permanent.

⁶This data limitation implies that some person-wave observations become unusable because the outcome does not exist in a relevant wave. For instance, consider someone who experiences a health shock in wave 10. We observe the probability of working past 70 for this person in the wave after their shock (wave 11), but not in the wave of the shock or earlier waves.

4.2 The Effects of Specific Health Conditions

Next, we unpack the findings on new diagnoses by studying the underlying conditions that make up the summary measure separately. For each specific condition, we analyze the impact of the shock on the probabilities of working past 62, 65, and 70. This exercise allows us to provide evidence on which diagnoses could be driving the overall effect. A disadvantage is that the sample sizes become (sometimes substantially) smaller.

We present graphs of the nonparametric event study results in Figure 3 for the probability of working past 65. We present results for 62 and 70 in Appendix Figures A.6 and A.7. Each graph corresponds to a different diagnosis. There are almost no pre-shock period estimates that are statistically different from zero. Indeed, for many of the diagnoses, the pattern of the pre-shock period estimates reveals encouragingly flat trends. However, there are a few shocks for which the pattern of the estimates suggests that the outcomes were trending downward even before the shock. For example, the probabilities of working past 62 and 65 appear to exhibit downward trends before heart disease diagnoses. Recall that the parametric event study regressions will account for these trends when quantifying the overall effects.

Some health conditions impact expectations about future work, whereas others appear to have little to no effect. For example, the graphs for arthritis, cancer, lung disease, and strokes in Figure 3 and Appendix Figure A.6 show some visually apparent declines in the likelihood of working past older ages after the diagnoses. In contrast, the graphs for diabetes, heart disease, and high blood pressure show point estimates that appear to evolve more smoothly. The results for psychiatric problems are perhaps less clear; the post-shock estimates are negative but they also appear to move more in accordance with the pre-shock trends.

Panel B of Table 2 presents the results from the parametric event study regressions. We find statistically significant declines in the likelihood of working past 62 for arthritis and cancer, and we find statistically significant declines in the likelihood of working past 65 for cancer and strokes. The estimates for lung disease suggest large declines but they are not statistically different from zero at conventional levels; the p-value for the estimates for working past 62 and 65 are 0.145 and 0.110, respectively. The estimates for high blood pressure and psychiatric problems are negative but are not statistically different from zero. In contrast, the point estimates for diabetes and heart disease would at face value suggest, if anything, weak evidence of updating expectations towards working longer. We emphasize that while some of the estimates that are not statistically significant are relatively small (like the point estimate for working past 65 after a diagnosis of high blood pressure), others are more sizable (like the estimates for psychiatric problems) but are not precisely estimated.

The magnitudes of the statistically significant estimates are large. The point estimate for the decline in the likelihood of working past 62 after an arthritis diagnosis is 4.9 percentage points, which corresponds to a 10.7% decrease when compared to the baseline mean. The results for cancer diagnoses indicate 7.0 and 6.0 percentage point declines in the likelihood of working past 62 and 65, which are 15.9% and 20.0% decreases when compared to the baseline means of 44% and 30%, respectively. Strikingly, the point estimate for the likelihood of working past 65 after a stroke corresponds to a 44.6% decline (although we note the sample size for this shock is very small).

One qualitative takeaway from these estimates is that the effects of adverse health shocks can be nuanced. Some diagnoses clearly impact retirement expectations, whereas the evidence for others is less clear. For instance, our findings highlight how arthritis, cancer, and strokes can lead to major changes in expectations about working longer. In contrast, the evidence for high blood pressure and diabetes suggests smaller, if any, effects. Of course, different health conditions may impact current work capacity, future work capacity, medical expenses, savings, and other factors differently. It could be that the often-chronic nature of arthritis contributes to larger effects for that diagnosis. Health conditions that are expected to worsen over time and last into the future may have stronger impacts on retirement expectations. It could also be that cancer has meaningful impacts on mortality expectations that cause people to update their plans towards retiring earlier. Other conditions, like diabetes, might have a less severe impact on work capacity while meaningfully increasing future medical expenses.

4.3 Robustness Checks

We conduct several robustness checks. For simplicity, we focus on our main health shocks and the probabilities of working past 62 and 65, although we also assess the robustness of our estimates for the specific health conditions and for the probability of working past 70 in the appendix (see Appendix Tables A.4, A.5, A.6, and A.7).

Table 3 presents the main robustness results. The columns of the table correspond to different health shocks and outcome variables. The rows of the table correspond to different robustness checks and indicate how the robustness specification differs from the baseline specification. Row A reproduces the baseline estimates for ease of comparison.

First, we assess the robustness of our results to standard regression specification checks in rows B through F. Row B includes control variables in the regression specification. Specifically, we include a variable for age and indicator variables for being female, for being white,

for having attended college, and for being married. This change has little effect on our estimates. The magnitudes of the estimates are similar, and the estimates that are statistically significant in the baseline specification remain statistically significant. Row C includes wave fixed effects instead of cohort-by-wave fixed effects. This change results in (i) estimates for health status declines that are larger than the baseline estimates and that are statistically significant at the 1-percent level, (ii) estimates for new diagnoses that are similar to the baseline estimates, and (iii) estimates for hospitalizations that are meaningfully larger than their baseline level.

Row D includes individual fixed effects. To include these fixed effects, we need to address the identification problem that arises because of the collinearity between individual fixed effects (which subsume time-of-shock fixed effects), time fixed effects, and time-relative-to-shock fixed effects, which are our parameters of interest (Dobkin et al., 2018; Borusyak, Jaravel and Spiess, 2024). To address this issue and obtain identification, we include individual fixed effects and multi-wave fixed effects instead of wave fixed effects, which effectively assumes that the macroeconomic conditions controlled for by time fixed effects evolve slowly.⁷ The results for health status declines and new diagnoses are similar to their baseline estimates. In contrast, the estimates for hospitalizations from this specification are larger in magnitude than their baseline counterparts, and they are statistically significant. In general, this approach also leads to more precise estimates.

Rows E and F assess the sensitivity of our estimates to the use of person-level sample weights. We do not use survey weights in our baseline specification because our goal is not to produce population-level statistics but is rather to provide causal estimates of health shocks for the subsample of people who experience the shocks. In this type of analysis, it is less clear whether one should use sample weights, although one practical reason to do so is to increase the precision of estimation (Solon, Haider and Wooldridge, 2015). Row E simply adds sample weights to the baseline specification. Our estimates are more sensitive to this choice than the other specification checks and are no longer statistically significant at conventional levels. The magnitudes are smaller, consistent with larger effects for people with smaller sample weights, although the estimates are also typically within the confidence intervals of the baseline estimates. Notably, the standard errors are substantially larger, in part because the procedure used to generate sample weights yields weights of zero for approximately 8% of our analysis samples, reducing the effective sample size. Thus, in an

⁷We have an odd number of waves, so we cannot use two-wave fixed effects; we modify this approach by grouping the last three waves into one “multi-wave” fixed effect.

attempt to gain precision, Row F uses survey weights and includes individual fixed effects. The standard errors, the point estimates, and the statistical significance of the estimates from this more-controlled specification are more similar to our baseline results.

Second, we assess the robustness of our results to sample construction choices and missing data in rows G through K. Because our sample sizes are not especially large, our baseline specification includes as much non-missing data as possible; we include all usable person-wave observations in the analysis samples between three waves before the shock and one wave after the shock. Row G takes a different approach by limiting the people in the analysis sample to only those who experience the shock of interest and have no missing values for any outcomes. The takeaways do not change when we look at this subsample. Row H takes an even more restrictive approach by focusing on a shorter, balanced panel. This analysis is limited to subsamples of observations that (i) are for people with non-missing values for all outcomes and (ii) are for survey waves corresponding to event times between $\tau = -2$ and $\tau = 0$, which are the waves for which everyone appears in the data by construction. The advantage of the balanced panels are that the sample composition is the same at all relative time periods. The disadvantage is that sample sizes are smaller. The magnitudes of the estimates are fairly similar to their baseline counterparts and the standard errors are larger.

Row I addresses missing data due to proxy interviews by setting the missing values to zero.⁸ We are effectively assuming that people who use a proxy respondent are unable to do the survey interview and also unable to work past later ages. The advantage of imputing these values is that it allows us to use more information contained in the data. The disadvantage is that it could overstate the effects of health shocks, if experiencing a health shock causes more proxy interviews and the actual probabilities of working past older ages are greater than zero. The resulting point estimates are similar to our baseline estimates, but slightly larger in magnitude.

Rows J and K address concerns related to Covid. For example, Covid introduced complexities in data collection efforts and led to disruptions in labor markets that may have influenced expectations about future work. First, Row J drops observations of individuals interviewed in March, April, and May of 2020 to mitigate the missing data concerns from skip pattern errors discussed earlier. The estimates are very similar, as few people in our analysis samples were interviewed during these months. Second, Row K drops all observations from waves 15 and 16, both of which could have been influenced by Covid in general. The results

⁸The RAND HRS Longitudinal File contains detailed codes for missing values. We replace values that are missing because the information is “not available due to skip patterns, typically because the interview is by proxy respondent” with zeros.

are reasonably similar to those from our baseline specification, although the estimates have larger standard errors, consistent with reduced sample sizes. The most notable difference is that the estimate for the probability of working past 62 after a decline in health status is smaller and is no longer statistically significant at conventional levels.

Finally, we assess the robustness of our main estimates to using a two-way fixed effects specification and one of the recently-developed methods that addresses issues with these models when treatment effects are heterogeneous (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021). We use the interaction weighted (IW) estimator developed by Sun and Abraham (2021) because they revisit a two-way fixed effects specification in Dobkin et al. (2018). We can therefore follow the implementation in their illustrative application closely. The basic idea is to estimate overall treatment effects by calculating the appropriately-weighted average of the underlying effects for each “cohort” of treated people (defined by the survey wave in which the person experiences the health shock of interest).⁹

Appendix Figure A.8 presents the IW estimates. The patterns are broadly similar to our main nonparametric estimates, although the pre-trends for working past 62 in the health status declines sample appear noisier and the pre-trends for both outcomes in the new diagnoses sample appear flatter. Each graph also displays the point estimate and standard error for the survey wave of the shock, which is the closest analog to our parametric estimates. Both of the key, wave-of-shock estimates for health status declines are statistically significant; the estimate for working past 62 is larger than the corresponding main estimate, whereas the estimate for working past 65 is quite similar to the main estimate. The results for new diagnoses are also similar to the main results. The magnitudes of both key estimates are slightly larger than their corresponding estimates in the main analysis, and the estimate for working past 62 is statistically different from zero. Finally, the lack of evidence for a response after hospitalizations is also similar to that in our main analysis.

Overall, these robustness checks bolster our confidence in our empirical approach and the baseline estimates. The size and statistical significance of the estimates for health status

⁹To implement their approach, we start with our main analysis samples as constructed above. We keep person-wave observations in an event window spanning four waves before the shock to one wave after the shock. We keep one more pre-period wave than in our primary analysis because we need to exclude two relative time period indicators to address multicollinearity, and because we want to estimate effects for more than one pre-shock wave. Following Sun and Abraham (2021), we exclude one wave prior to the shock and four waves prior to the shock. We then estimate dynamic treatment effects for each cohort who experiences the health shock of interest between waves 10 and 15 and calculate the average of these effects, where the weights are the estimated cohort shares. Note that we cannot estimate the effects for those treated in wave 16 because this estimator uses the final treated group as a control group, which also means that we must exclude observations in wave 16, when the final cohort is treated, from the sample contributing to the estimates.

declines and new diagnoses are reasonably stable across specifications, samples, and approaches to estimation. The estimates for hospitalizations are more sensitive to specification choices; our baseline specification indicates a lack of evidence of responses, but some of the alternative specifications produce estimates that are larger in magnitude and statistically significant.

4.4 Assessing the Importance of Avoiding Sample Selection Issues

An important feature of our study that supports causal interpretations of the estimates is the more recent survey data. The earlier waves of the HRS contain retirement expectations outcomes only for workers. In contrast, we leverage updated data that contain these outcomes even for non-workers. We take two approaches to assess the importance of overcoming the sample selection issues that the earlier waves of the data presented.

First, we redo our analysis using only the earlier waves instead of only the later waves. Specifically, in our main analysis, one of our initial sample construction steps involves limiting the underlying data to only waves 8 through 16. Here, we limit the underlying data to only waves 1 through 7. We then define the main health shocks of interest, construct the analysis samples, and estimate regressions in the same way as our main analysis.

Appendix Figure A.9 presents the nonparametric event study results, and panel A of Table 4 presents the parametric results. These results from the earlier sample waves starkly contrast with our main results. Overall, the patterns of the nonparametric estimates indicate either a lack of evidence of a response or, if anything, an increase in the likelihood of working at later ages. Indeed, the parametric estimates indicate no statistically significant evidence of an effect for health status declines or hospitalizations but strong evidence of an increase in the likelihood of working longer after a new diagnosis.

Second, we redo our analysis using the later survey waves but restrict the sample to include only workers. Specifically, we use the same underlying samples of people in our main analysis, but we keep only the person-wave observations for which the person reports working full time or part time. In contrast to the first approach, which generates a different sample of people (those who experience the health shocks of interest in earlier waves), this second approach has the advantage of keeping the people in our sample fixed while still allowing us to look at the impact that selection would have on our estimates if the survey did not begin recording responses for non-workers.

Appendix Figure A.10 presents the nonparametric event study results, and panel B of Table 4 presents the parametric results. When we use this selected sample, we find no

evidence that health shocks impact retirement expectations. Because the outcome variables ask about the probability of working full time, we also conduct the same analysis, but where we keep only person-wave observations of full-time workers. Panel C of the table presents these estimates, which are similar to those for all workers.

Overall, the results from both approaches are consistent with the idea that people who drop out of the selected sample because they (temporarily or permanently) stop working also believe they are less likely to work past 62 and 65. This evidence is thus consistent with upward bias in prior studies limited to only the restricted sample of people who are still working, who are likely to be healthier and to have greater work capacity than the full sample. This upward bias would lead to estimates that understate the true declines in expectations from health shocks. Avoiding sample selection is thus important for assessing the impact of health on retirement expectations, as we would have reached different conclusions if we could not use the updated data that cover workers and non-workers.

5 Discussion and Conclusion

This paper provides new causal evidence on how health shocks affect retirement expectations. The results are relevant for policymakers and practitioners concerned with assessing the retirement income security of older Americans.

First, our results highlight how health risks can cause people to expect to retire earlier than they otherwise would have. This type of earlier retirement may lead to greater reliance on social insurance programs and public assistance in general. To the extent that public systems do not adequately insure against these health risks, there could be important implications for financial security in retirement. For example, Social Security and Medicare are two of the largest public programs in the U.S. and provide access to old-age benefits and health insurance starting at 62 and 65, respectively. Declines in health status can reduce overall lifetime earnings and lead individuals to retire before becoming eligible for benefits, highlighting a potentially important role for the Disability Insurance program. Moreover, reductions in lifetime earnings often reduce benefits that individuals receive.

Second, our results are informative for retirement plan design and the administration of tax-advantaged retirement savings accounts. The expected retirement date is a key input into optimal strategies within a savings plan. Indeed, target date funds explicitly anchor investments to expected retirement dates and are an increasingly important part of savings portfolios (Shoven and Walton, 2021). Our results highlight the importance of health in

shaping retirement timing uncertainty and suggest that people may value flexibility in savings schemes like the ability to update their elected expected retirement age. Additionally, our analysis points to specific shocks that may require changes to savings plans.

Finally, our analysis advances our understanding of how people manage uncertain events in preparation for retirement. People may modify their plans by, for instance, working longer, changing the amount they work while working, saving more, or consuming less. While Bronshtein et al. (2019) highlight the power of working longer for retirement readiness, our analysis implies that extending employment may not always be an available option. In these cases, savings responses may be critical. However, evidence shows that retirement savings often evolve passively (Madrian and Shea, 2001; Chetty et al., 2014) and that retirement savings decisions are strongly linked to labor supply decisions (García-Miralles and Leganza, 2024). This passivity raises concerns that many may not optimally update savings in response to events that change expected retirement dates. While assessing optimal savings responses to health shocks is beyond the goal of this paper, studying changes in retirement expectations is an essential first step in understanding whether people should adjust savings, the extent to which they should do so, and how financially secure they will be in retirement.

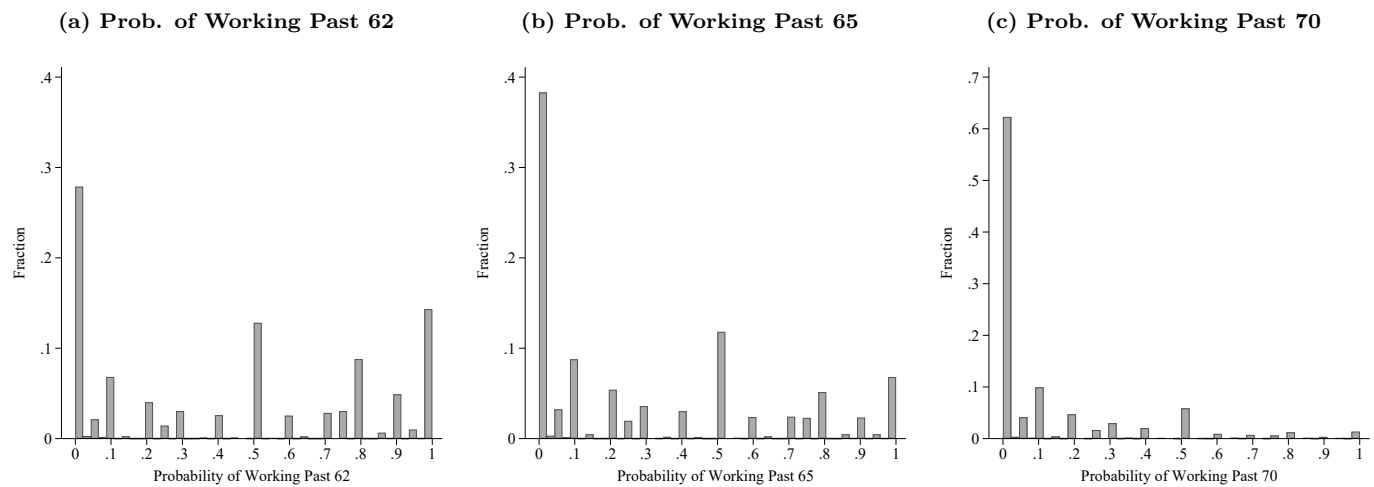
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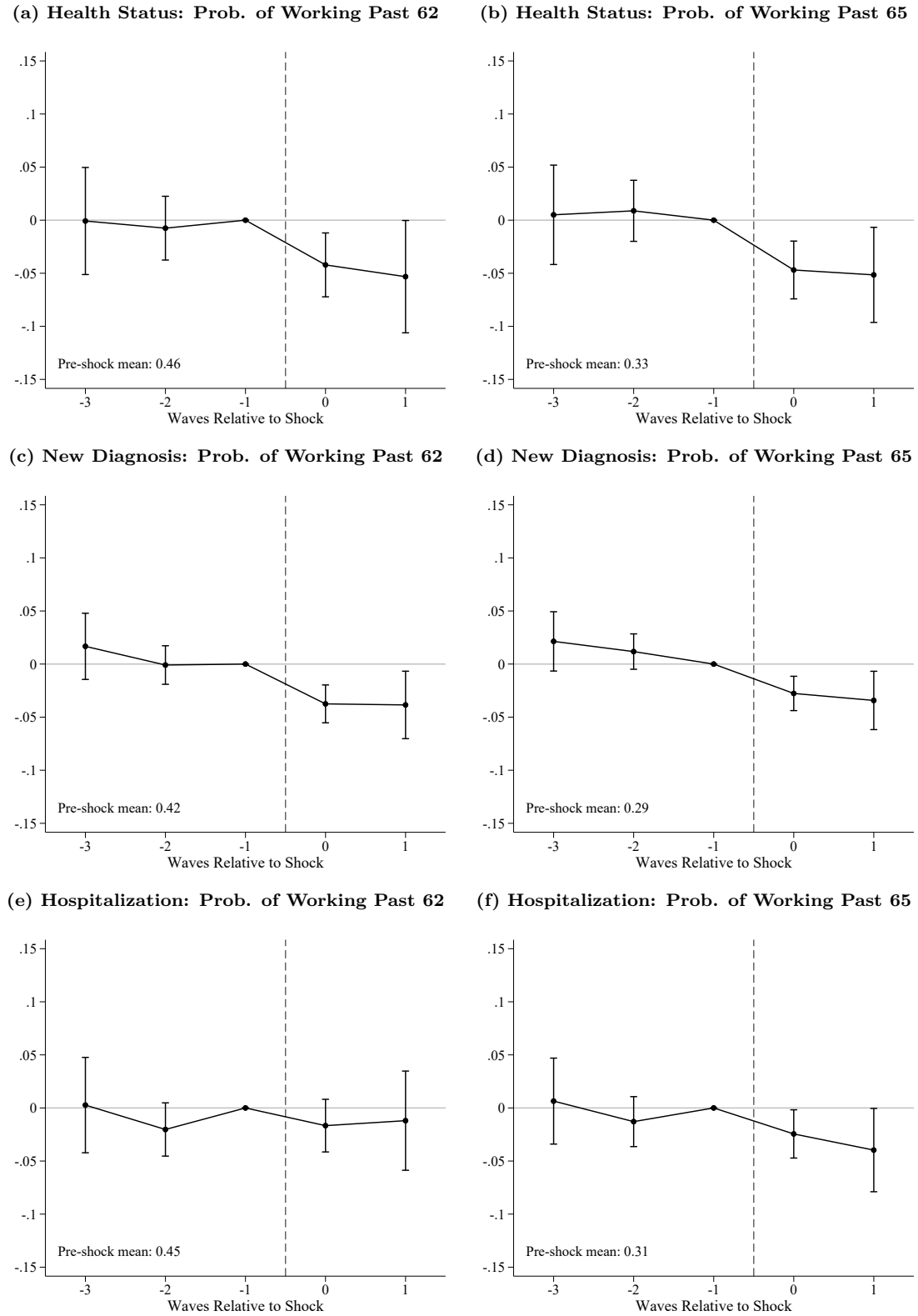
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Figure 1: Histograms of Retirement Expectation Outcome Variables



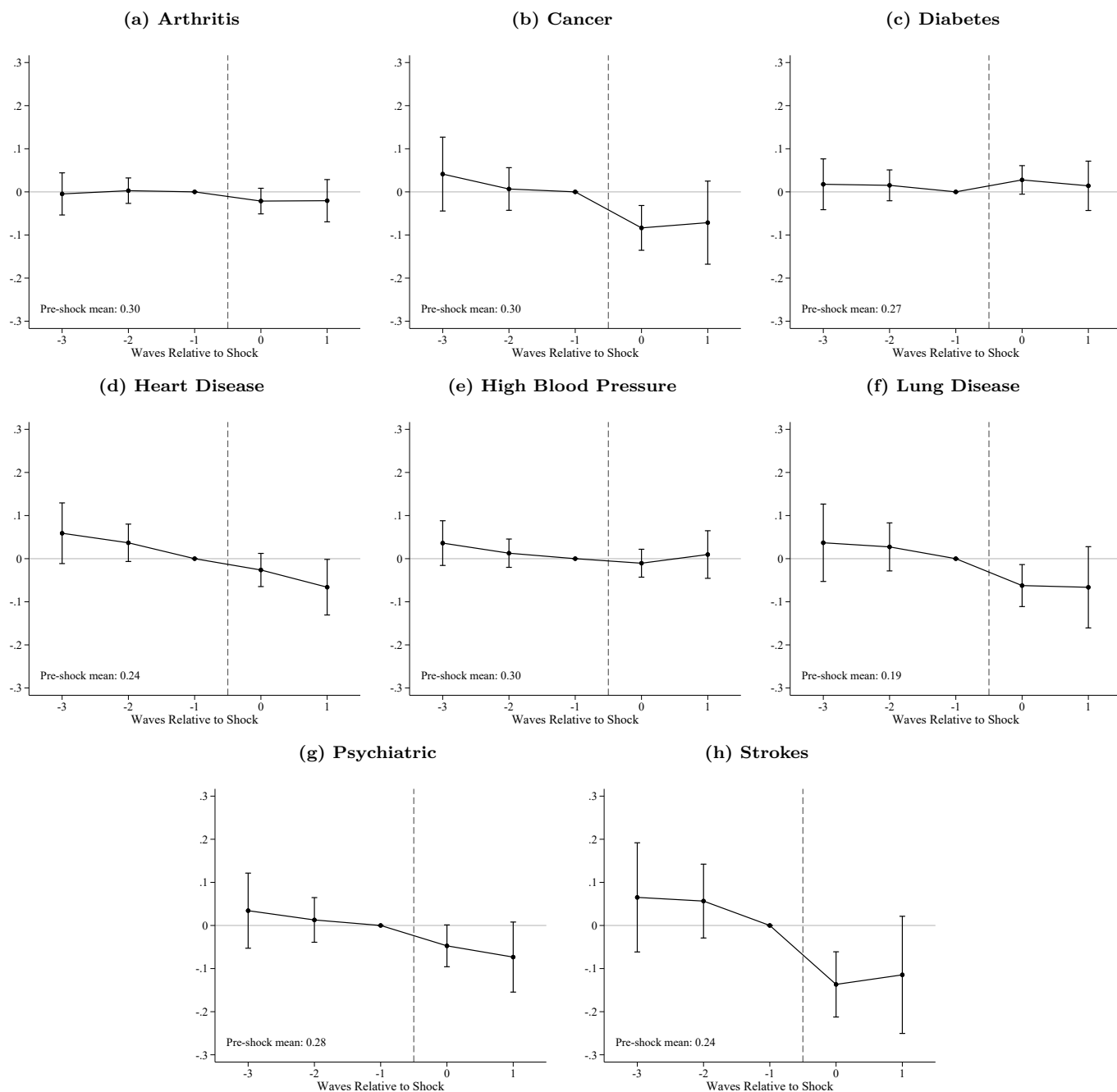
Notes: This figure presents histograms of our retirement expectation outcome variables. The underlying sample consists of people in the HRS data between ages 50 and 60 surveyed in waves 8 through 16. Each graph corresponds to a different outcome variable of interest.

Figure 2: Effects of Main Health Shocks on Retirement Expectations



Notes: This figure presents the nonparametric event study estimates for our three main health shocks (declines in health status, new diagnoses, and hospitalizations) and our two main outcome variables (the probabilities of working past 62 and 65). The graphs plot point estimates and confidence intervals for the δ_T coefficients in equation (1).

Figure 3: Effects of Specific Health Conditions on the Probability of Working Past 65



Notes: This figure presents the nonparametric event study estimates for each of the eight specific health conditions that we study and for the probability of working past 65. The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Table 1: Summary Statistics in the Survey Wave Before Each Main Health Shock

	Health Status		New Diagnosis		Hospitalization	
	Mean (1)	Obs. (2)	Mean (3)	Obs. (4)	Mean (5)	Obs. (6)
Age	55.73	974	55.59	2,904	55.83	1,370
Male	0.43	974	0.39	2,904	0.43	1,370
White	0.56	974	0.58	2,904	0.63	1,370
Married	0.64	974	0.66	2,904	0.67	1,370
Any College	0.48	974	0.52	2,904	0.53	1,370
Probability of Working Past 62	0.46	921	0.42	2,784	0.45	1,324
Probability of Working Past 65	0.33	910	0.29	2,745	0.31	1,304
Probability of Working Past 70	0.13	727	0.11	2,153	0.12	968

Notes: This table reports summary statistics for the samples of people who experience our main health shocks of interest. For each analysis sample, the underlying data consist of observations of people between ages 50 and 60 in survey waves 8 through 16 who experience the corresponding health shock. The sample means and observations presented in the table are for the survey wave before the shock.

Table 2: The Effect of Health Shocks on Retirement Expectations

	Probability of Working Past 62				Probability of Working Past 65			
	Estimate (1)	Mean (2)	Clusters (3)	Obs. (4)	Estimate (5)	Mean (6)	Clusters (7)	Obs. (8)
A: Main Shocks								
Health Status Decline	-0.041** (0.018)	0.46	966	3,635	-0.045*** (0.016)	0.33	966	3,589
New Diagnosis	-0.028*** (0.010)	0.42	2,879	10,758	-0.017* (0.010)	0.29	2,880	10,643
Hospitalization	-0.012 (0.014)	0.45	1,359	5,118	-0.019 (0.013)	0.31	1,359	5,070
B: Specific Diagnoses								
Arthritis	-0.049** (0.019)	0.46	900	3,402	-0.024 (0.017)	0.30	901	3,369
Cancer	-0.070** (0.033)	0.44	255	964	-0.060** (0.030)	0.30	255	953
Diabetes	0.024 (0.021)	0.41	595	2,257	0.036* (0.020)	0.27	595	2,219
Heart Disease	0.046* (0.026)	0.36	397	1,469	0.002 (0.025)	0.24	397	1,459
High Blood Pressure	-0.023 (0.020)	0.45	770	2,856	0.008 (0.019)	0.30	770	2,822
Lung Disease	-0.051 (0.035)	0.29	228	840	-0.045 (0.028)	0.19	228	830
Psychiatric	-0.043 (0.032)	0.41	356	1,334	-0.030 (0.030)	0.28	356	1,325
Stroke	-0.072 (0.045)	0.33	145	515	-0.107** (0.043)	0.24	145	506

Notes: This table reports the parametric event study estimates for the probabilities of working past 62 and 65. Panel A presents estimates for our main health shocks of interest (declines in health status, new diagnoses, and hospitalizations). Panel B presents estimates for specific health conditions. Columns (1) and (5) present estimates of δ_0 from estimating equation (2). Standard errors clustered at the individual level are in parentheses. Columns (2) and (6) present the dependent variable mean in the survey wave before the health shock. Columns (3) and (7) present the number of clusters. Columns (4) and (8) present the number of observations.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Robustness of Main Estimates

	Health Status		New Diagnosis		Hospitalization	
	Age 62 (1)	Age 65 (2)	Age 62 (3)	Age 65 (4)	Age 62 (5)	Age 65 (6)
A. Baseline Specification	-0.041** (0.018)	-0.045*** (0.016)	-0.028*** (0.010)	-0.017* (0.010)	-0.012 (0.014)	-0.019 (0.013)
B. Include Control Variables	-0.038** (0.018)	-0.041** (0.016)	-0.029*** (0.010)	-0.018* (0.010)	-0.014 (0.014)	-0.020 (0.013)
C. Include Wave Fixed Effects	-0.054*** (0.017)	-0.058*** (0.016)	-0.028*** (0.010)	-0.018* (0.010)	-0.026* (0.014)	-0.030** (0.013)
D. Include Individual Fixed Effects	-0.047*** (0.015)	-0.041*** (0.014)	-0.033*** (0.009)	-0.020** (0.008)	-0.033*** (0.012)	-0.032*** (0.012)
E. Use Survey Weights	-0.025 (0.028)	-0.039 (0.025)	-0.012 (0.016)	0.002 (0.015)	0.001 (0.021)	0.010 (0.021)
F. Survey Weights + Individual Fixed Effects	-0.039* (0.023)	-0.040* (0.020)	-0.026** (0.013)	-0.006 (0.013)	-0.028 (0.019)	-0.008 (0.018)
G. Subsample: No Missing Values	-0.037* (0.019)	-0.041** (0.017)	-0.027** (0.011)	-0.018* (0.010)	-0.016 (0.015)	-0.019 (0.013)
H. Subsample: Balanced Panel	-0.046** (0.022)	-0.032 (0.021)	-0.038*** (0.013)	-0.017 (0.012)	-0.028* (0.016)	-0.025 (0.016)
I. Impute Proxy Interview Values	-0.044** (0.018)	-0.047*** (0.016)	-0.030*** (0.010)	-0.018* (0.010)	-0.018 (0.014)	-0.022* (0.013)
J. Exclude Early 2020 Interviews	-0.040** (0.018)	-0.044*** (0.017)	-0.021* (0.011)	-0.013 (0.010)	-0.012 (0.015)	-0.020 (0.013)
K. Exclude Covid Waves	-0.028 (0.020)	-0.032* (0.019)	-0.021* (0.012)	-0.021* (0.011)	-0.005 (0.017)	-0.018 (0.015)

Notes: This table reports results from assessing the robustness of our main estimates, which correspond to our main health shocks (declining health status, new diagnoses, and hospitalizations) and our main outcome variables (the probabilities of working past 62 and 65). The estimates presented are the δ_0 s from estimating equation (2). Each column corresponds to a different health shock and outcome variable. Each row corresponds to a different robustness check. Row A reproduces the baseline estimates for ease of comparison. Row B adds control variables to the regressions. Row C uses wave fixed effects instead of HRS-cohort-by-wave fixed effects. Row D uses individual fixed effects and multi-wave fixed effects. Row E uses survey weights when estimating the regressions. Row F uses survey weights and includes individual fixed effects and multi-wave fixed effects. Row G focuses on a subsample of people who never have missing values for any of the outcome variables. Row H focuses on the balanced panel of observations of individuals without missing values in survey waves corresponding to event times between $\tau = -2$ and $\tau = 0$. Row I includes observations from proxy interviews by replacing the missing values with zeros. Row J excludes observations of people interviewed in March, April, or May of 2020. Row K excludes all observations in survey waves 15 and 16.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4: Assessing the Importance of Avoiding Sample Selection Issues

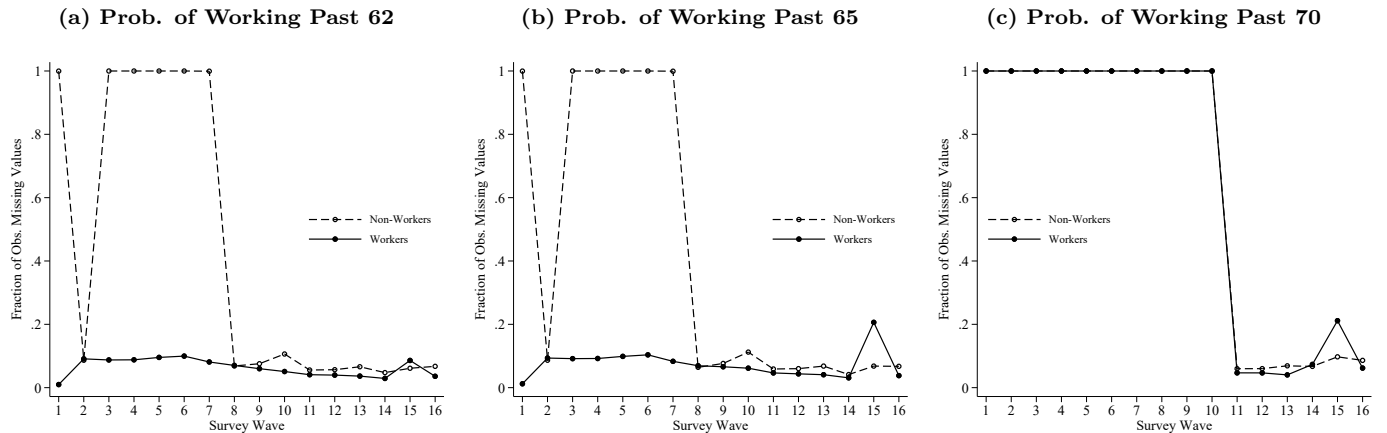
	Probability of Working Past 62				Probability of Working Past 65			
	Estimate (1)	Mean (2)	Clusters (3)	Obs. (4)	Estimate (5)	Mean (6)	Clusters (7)	Obs. (8)
A: Earlier Waves								
Health Status Decline	0.000 (0.026)	0.59	902	2,591	-0.019 (0.021)	0.24	900	2,582
New Diagnosis	0.043*** (0.015)	0.57	2,218	6,523	0.016 (0.013)	0.22	2,215	6,498
Hospitalization	0.010 (0.019)	0.61	1,467	4,290	0.005 (0.015)	0.22	1,467	4,280
B: Workers in Later Waves								
Health Status Decline	-0.022 (0.024)	0.59	756	2,270	-0.029 (0.023)	0.41	755	2,231
New Diagnosis	0.004 (0.014)	0.57	2,099	6,460	0.012 (0.014)	0.39	2,096	6,357
Hospitalization	0.019 (0.020)	0.61	998	3,047	-0.009 (0.020)	0.42	996	3,003
C: Full-Time Workers in Later Waves								
Health Status Decline	-0.011 (0.027)	0.61	674	1,831	-0.033 (0.027)	0.43	673	1,797
New Diagnosis	-0.004 (0.016)	0.60	1,868	5,305	0.009 (0.015)	0.40	1,863	5,218
Hospitalization	0.012 (0.022)	0.64	886	2,518	-0.023 (0.022)	0.43	885	2,480

Notes: This table reports the parametric event study estimates for the probabilities of working past 62 and 65 when we study our main health shocks using alternative samples that are subject to sample selection concerns. Panel A presents estimates for our main health shocks of interest when we use earlier survey waves instead of later survey waves. Panel B presents estimates for our main health shocks of interest when we restrict our main analysis samples to workers instead of studying workers and non-workers. Panel C Presents estimates for our main health shocks of interest when we restrict our main analysis samples to full-time workers only. Columns (1) through (4) correspond to the probability of working past 62, and columns (5) through (8) correspond to the probability of working past 65. Columns (1) and (5) present estimates of δ_0 from estimating equation (2). Standard errors clustered at the individual level are in parentheses. Columns (2) and (6) present the dependent variable means in the survey wave before the health shock. Columns (3) and (7) present the number of clusters. Columns (4) and (8) present the number of observations.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

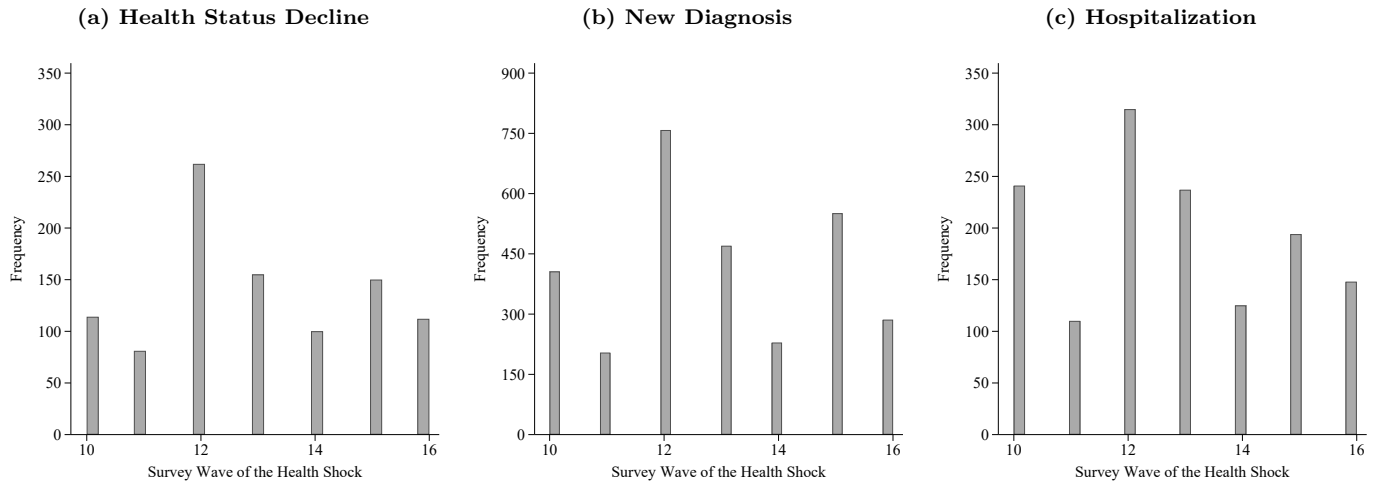
Appendix A Additional Figures and Tables

Figure A.1: Missing Data by Work Status Across Survey Waves



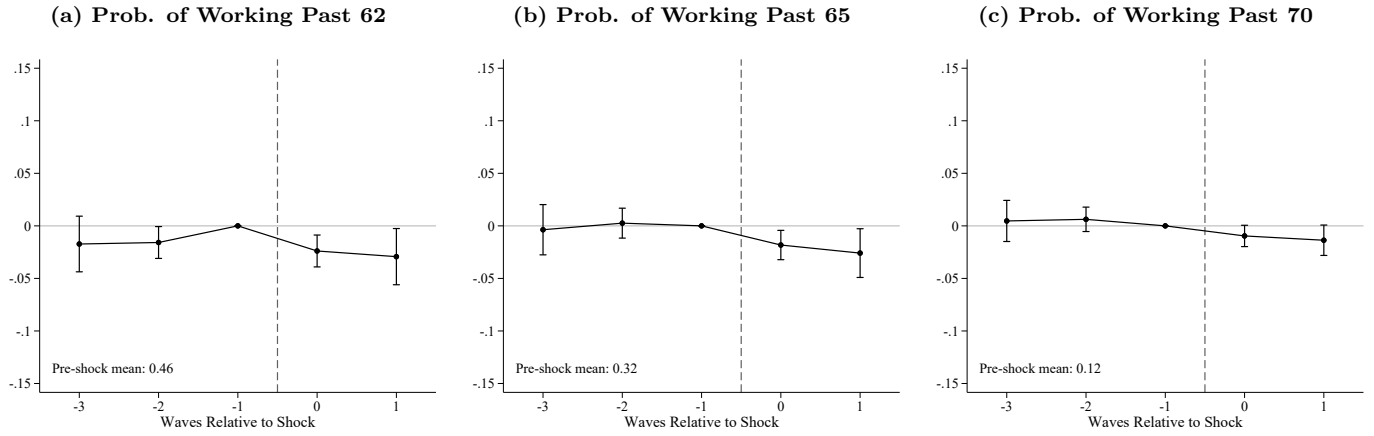
Notes: This figure illustrates the missing data issues with the earlier sample waves. The underlying sample consists of people in the HRS data between ages 50 and 60. Each graph corresponds to a different outcome variable of interest and plots the fraction of observations that have missing values for that outcome variable across survey waves for workers and non-workers separately. Graphs (a) and (b) show how the probabilities of working past 62 and 65 are missing for all non-workers in the earlier survey waves (except for wave 2); the rates of missing data for non-workers then decline to rates similar for workers starting with wave 8, which is when the underlying survey questions started being consistently asked to workers and non-workers. Graph (c) shows how the probability of working past 70 is missing for workers and non-workers before wave 11, before the outcome variable existed.

Figure A.2: Histograms of Health Event Waves



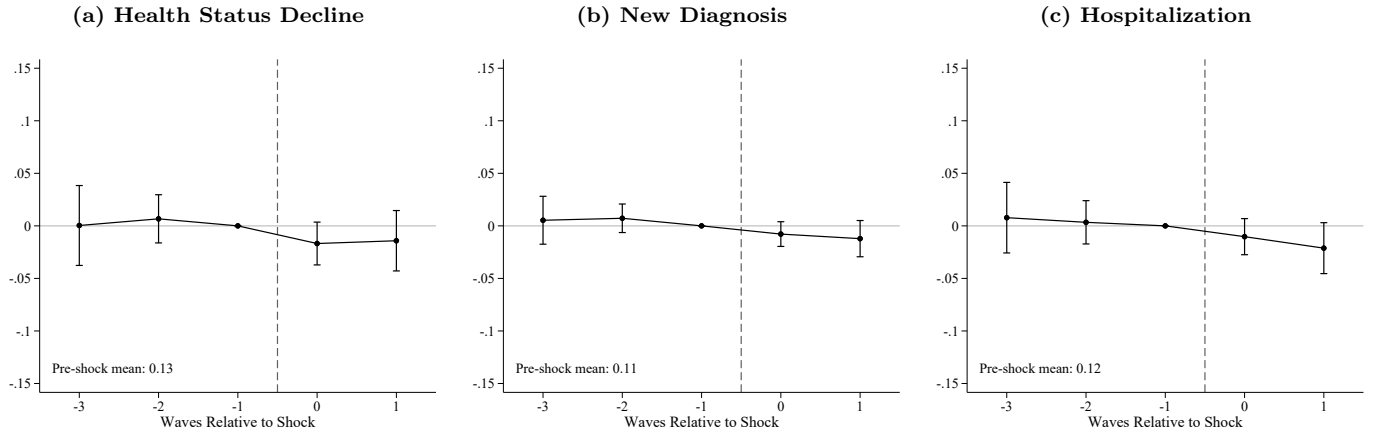
Notes: This figure presents histograms of the event survey waves for each of our main analysis samples. Each graph corresponds to a different health shock of interest and plots how many people experience that health shock in each survey wave.

Figure A.3: Effects of Any Decline in Subjective Health



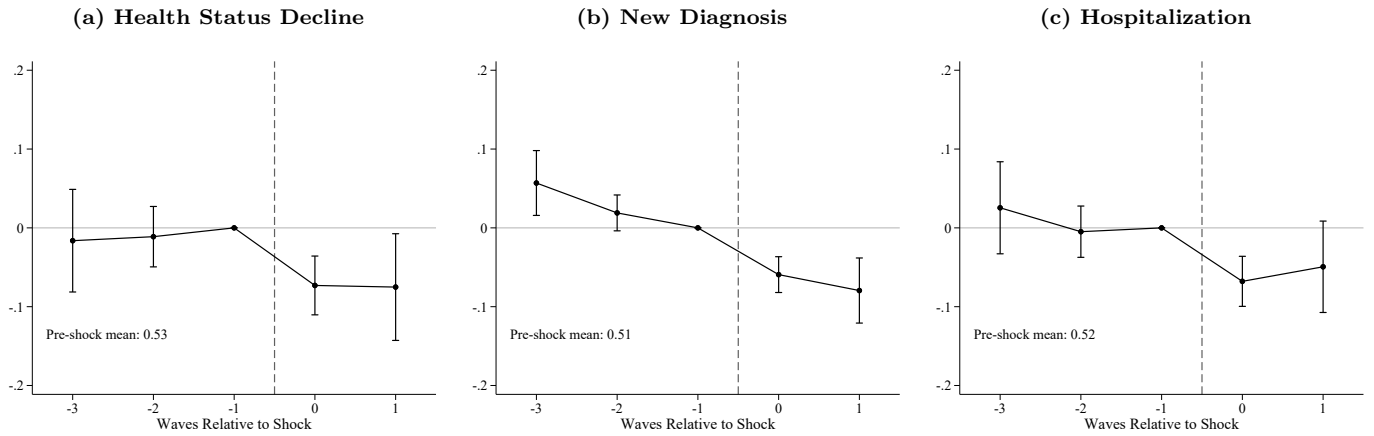
Notes: This figure presents the nonparametric event study estimates for the probability of working past 62, 65, and 70 for an alternative analysis sample of people experiencing health status declines. The sample consists of people who experience any downgrade in health status, instead of a decline from excellent, very good, or good to either fair or poor. The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Figure A.4: Effects of Main Shocks on the Probability of Working Past 70



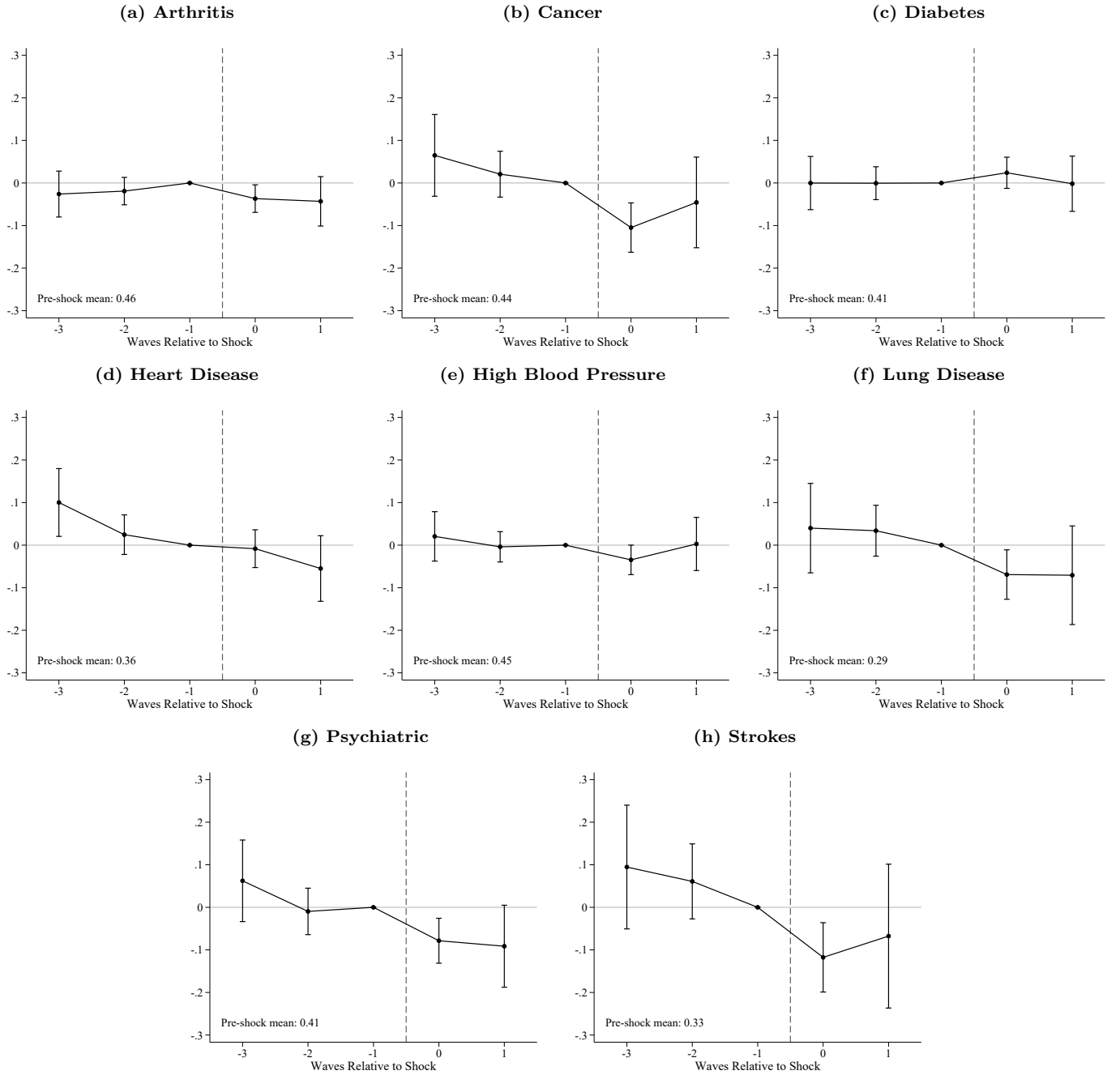
Notes: This figure presents the nonparametric event study estimates for our three main health shocks (declines in health status, new diagnoses, and hospitalizations) and for the probability of working past 70. The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Figure A.5: Effects of Main Health Shocks on Current Labor Supply



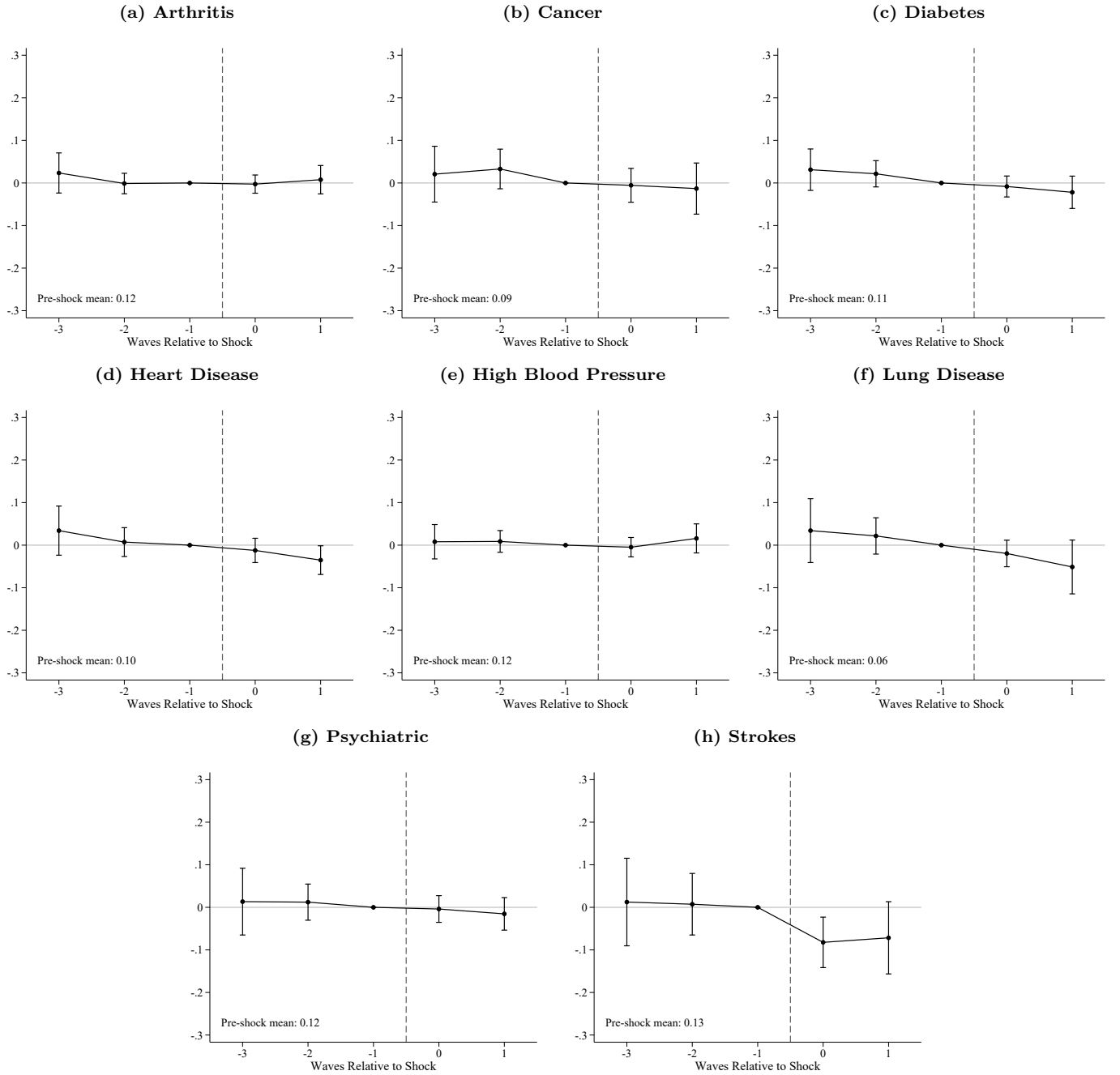
Notes: This figure presents the nonparametric event study estimates for our three main health shocks (declines in health status, new diagnoses, and hospitalizations) and for an outcome that captures current labor supply (an indicator variable for currently working full-time). The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Figure A.6: Effects of Specific Health Conditions on the Probability of Working Past 62



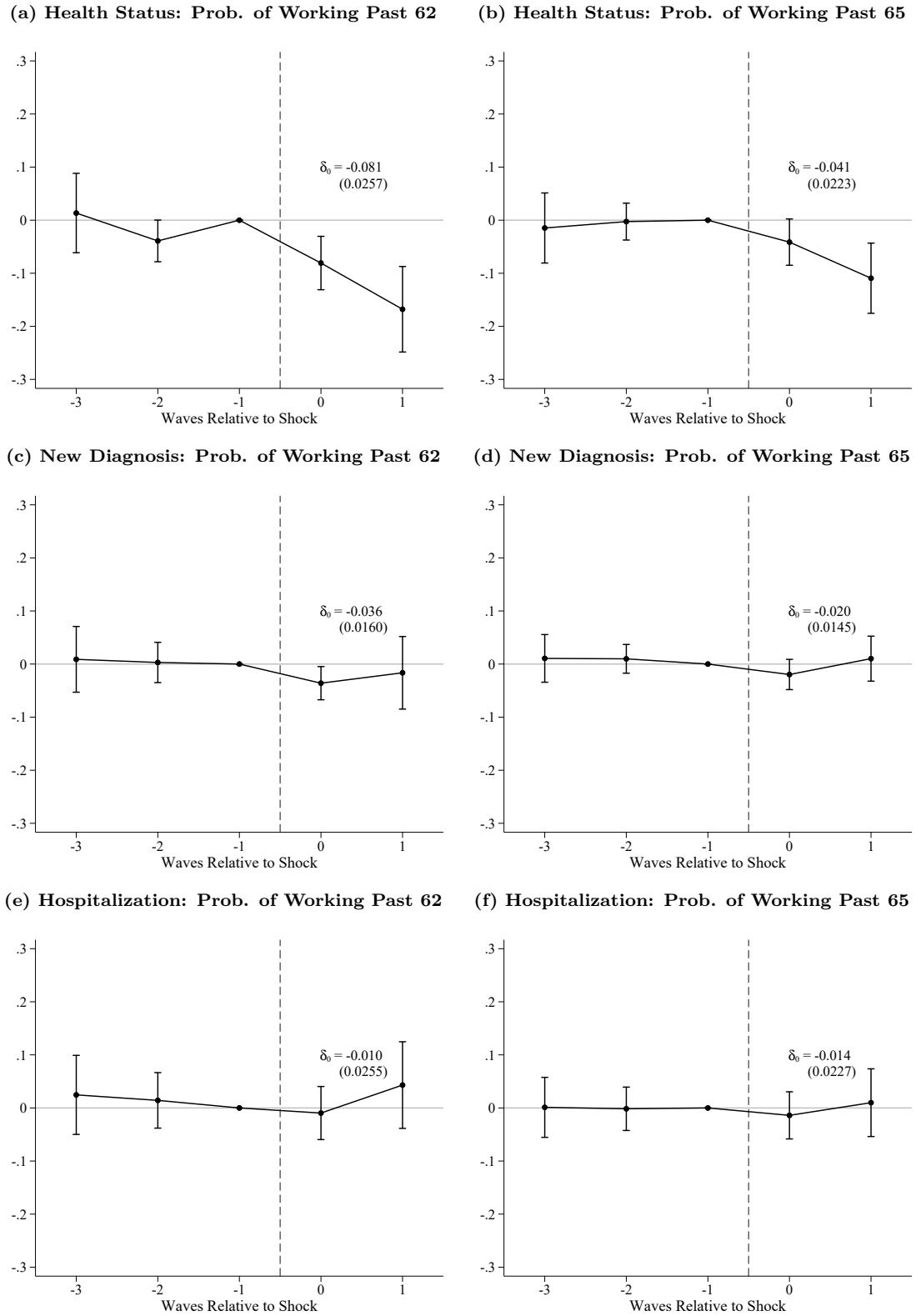
Notes: This figure presents the nonparametric event study estimates for each of the eight specific health conditions that we study and for the probability of working past 62. The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Figure A.7: Effects of Specific Health Conditions on the Probability of Working Past 70



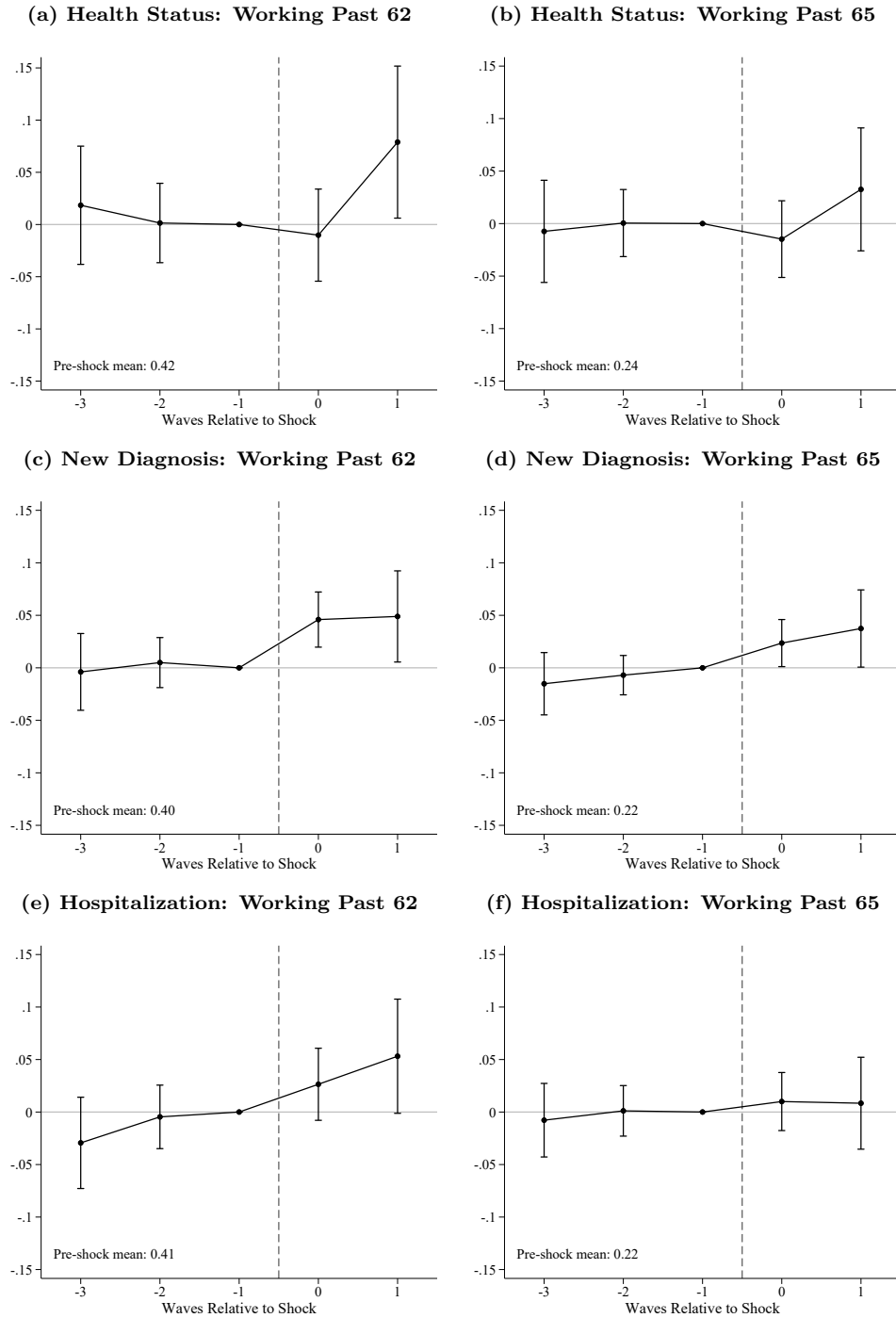
Notes: This figure presents the nonparametric event study estimates for each of the eight specific health conditions that we study and for the probability of working past 70. The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Figure A.8: Effects of Main Health Shocks: Interaction Weighted Event Study Estimates



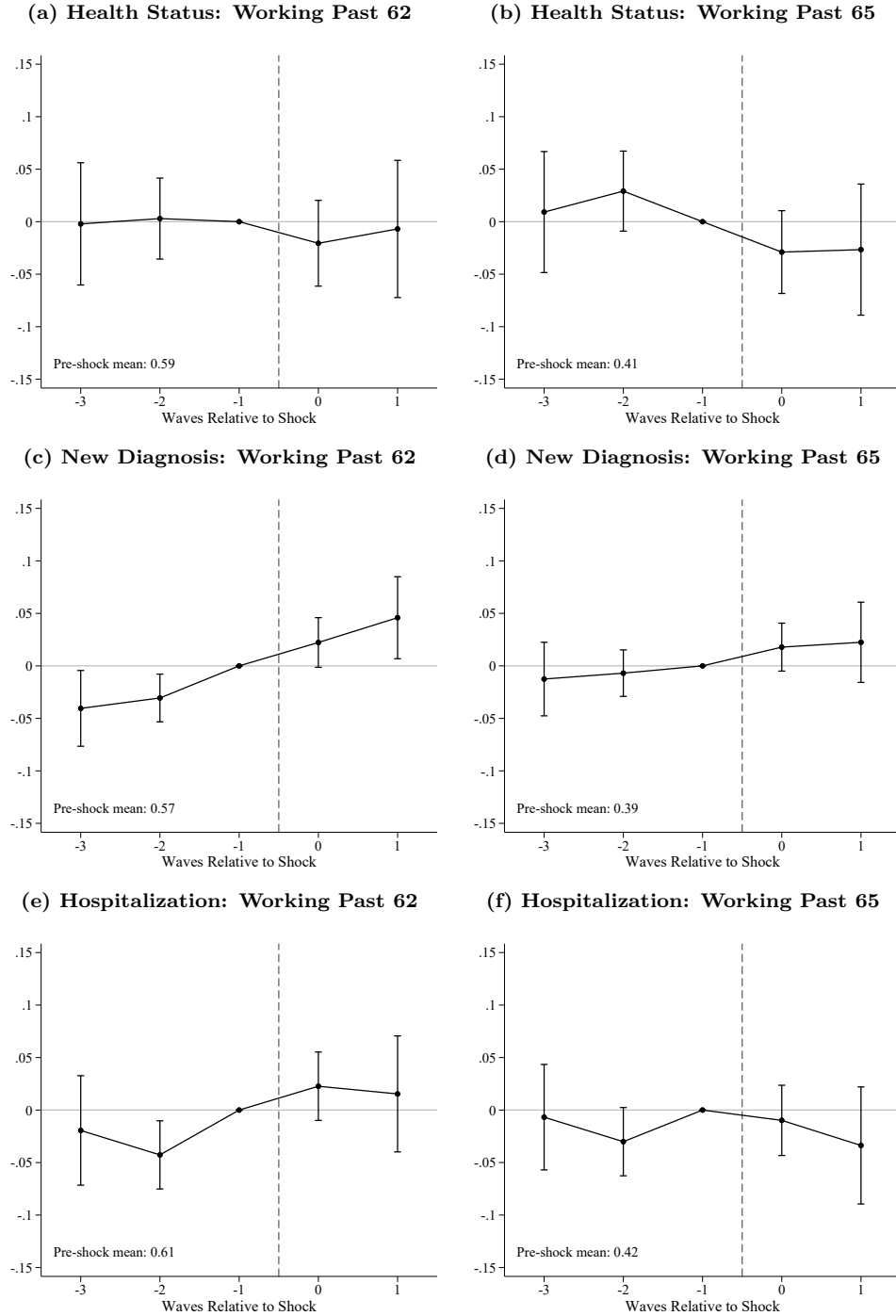
Notes: This figure presents event study estimates for our three main health shocks (declines in health status, new diagnoses, and hospitalizations) and our three main outcome variables (the probabilities of working past 62, 65, and 70) when we use the estimation approach from Sun and Abraham (2021).

Figure A.9: Assessing the Importance of Avoiding Sample Selection Issues: Results Using Earlier Survey Waves



Notes: This figure presents the nonparametric event study estimates for our three main health shocks (declines in health status, new diagnoses, and hospitalizations) and our two main outcome variables (the probabilities of working past 62 and 65) when we use earlier sample waves, which are subject to sample selection concerns, instead of later sample waves. The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Figure A.10: Assessing the Importance of Avoiding Sample Selection Issues: Results Using Workers in Later Survey Waves



Notes: This figure presents the nonparametric event study estimates for our three main health shocks (declines in health status, new diagnoses, and hospitalizations) and our two main outcome variables (the probabilities of working past 62 and 65) when we restrict our analysis samples to workers, which creates sample selection concerns, instead of studying workers and non-workers. The graphs plot point estimates and confidence intervals for the δ_τ coefficients in equation (1).

Table A.1: The Effect of Any Decline in Subjective Health Status

	Probability of Working Past 62 (1)	Probability of Working Past 65 (2)	Probability of Working Past 70 (3)
Estimate	-0.032*** (0.009)	-0.021** (0.008)	-0.007 (0.007)
Mean	0.46	0.32	0.12
Clusters	3,988	3,990	3,746
Observations	14,943	14,788	10,647

Notes: This table reports the parametric event study estimates for the probability of working past 62, 65, and 70 for an alternative analysis sample of people experiencing health status declines. The sample consists of people who experience any downgrade in health status, instead of a decline from excellent, very good, or good to either fair or poor. Standard errors clustered at the individual level are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.2: The Effect of Health Shocks on the Probability of Working Past 70

	Probability of Working Past 70			
	Estimate (1)	Mean (2)	Clusters (3)	Obs. (4)
A: Main Shocks				
Health Status Decline	-0.016 (0.013)	0.13	904	2,636
New Diagnosis	-0.005 (0.008)	0.11	2,670	7,743
Hospitalization	-0.006 (0.011)	0.12	1,236	3,533
B: Specific Diagnoses				
Arthritis	0.008 (0.014)	0.12	840	2,491
Cancer	0.006 (0.024)	0.09	236	682
Diabetes	0.008 (0.017)	0.11	567	1,714
Heart Disease	0.004 (0.021)	0.10	362	1,091
High Blood Pressure	-0.000 (0.015)	0.12	699	1,997
Lung Disease	-0.002 (0.024)	0.06	213	608
Psychiatric	0.004 (0.024)	0.12	334	968
Stroke	-0.076** (0.033)	0.13	136	377

Notes: This table reports the parametric event study estimates for the probability of working past 70. Panel A presents estimates for our main health shocks of interest (declines in health status, new diagnoses, and hospitalizations). Panel B presents estimates for specific health conditions. Column (1) present estimates of δ_0 from estimating equation (2). Standard errors clustered at the individual level are in parentheses. Column (2) presents the dependent variable mean in the survey wave before the health shock. Column (3) presents the number of clusters. Column (4) presents the number of observations.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.3: The Effects of Main Health Shocks on Current Labor Supply

	Working Full Time			
	Estimate (1)	Mean (2)	Clusters (3)	Obs. (4)
Health Status Decline	-0.081*** (0.023)	0.53	974	3,822
New Diagnosis	-0.030** (0.013)	0.51	2,903	11,253
Hospitalization	-0.053*** (0.018)	0.52	1,370	5,326

Notes: This table reports the parametric event study estimates for our main health shocks of interest (declines in health status, new diagnoses, and hospitalizations) and for an outcome that captures current labor supply (an indicator variable for currently working full-time). Column (1) present estimates of δ_0 from estimating equation (2). Standard errors clustered at the individual level are in parentheses. Column (2) presents the dependent variable mean in the survey wave before the health shock. Column (3) presents the number of clusters. Column (4) presents the number of observations.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.4: Robustness of Age 70 Estimates for the Main Health Shocks

	Health Status	New Diagnosis	Hospitalization
	Age 70 (1)	Age 70 (2)	Age 70 (3)
A. Baseline Specification	-0.016 (0.013)	-0.005 (0.008)	-0.006 (0.011)
B. Include Control Variables	-0.014 (0.013)	-0.006 (0.008)	-0.006 (0.011)
C. Include Wave Fixed Effects	-0.022* (0.013)	-0.010 (0.008)	-0.015 (0.011)
D. Include Individual Fixed Effects	-0.006 (0.012)	-0.006 (0.007)	-0.006 (0.011)
E. Use Survey Weights	-0.025 (0.021)	0.011 (0.012)	0.013 (0.017)
F. Survey Weights + Individual Fixed Effects	-0.021 (0.018)	0.005 (0.010)	0.013 (0.015)
G. Subsample: No Missing Values	-0.017 (0.014)	-0.008 (0.008)	-0.011 (0.012)
H. Subsample: Balanced Panel	-0.011 (0.017)	-0.003 (0.010)	-0.010 (0.014)
I. Impute Proxy Interview Values	-0.017 (0.013)	-0.005 (0.008)	-0.007 (0.011)
J. Exclude Early 2020 Interviews	-0.017 (0.014)	-0.002 (0.008)	-0.005 (0.011)
K. Exclude Covid Waves	-0.006 (0.015)	-0.006 (0.009)	-0.003 (0.013)

Notes: This table reports results from assessing the robustness of our estimates for the main health shocks (declining health status, new diagnoses, and hospitalizations) and for the probability of working past 70. The estimates presented are the δ_0 s from estimating equation (2). Each column corresponds to a different health shock. Each row corresponds to a different robustness check. Row A reproduces the baseline estimates for ease of comparison. Row B adds control variables to the regressions. Row C uses wave fixed effects instead of HRS-cohort-by-wave fixed effects. Row D uses individual fixed effects and multi-wave fixed effects. Row E uses survey weights when estimating the regressions. Row F uses survey weights and includes individual fixed effects and multi-wave fixed effects. Row G focuses on a subsample of people who never have missing values for any of the outcome variables. Row H focuses on the balanced panel of observations of individuals without missing values in survey waves corresponding to event times between $\tau = -2$ and $\tau = 0$. Row I includes observations from proxy interviews by replacing the missing values with zeros. Row J excludes observations of people interviewed in March, April, or May of 2020. Row K excludes all observations in survey waves 15 and 16.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.5: Robustness of Age 62 Estimates for Specific Health Conditions

	Arthritis (1)	Cancer (2)	Diabetes (3)	Heart Disease (4)	Blood Pressure (5)	Lung Disease (6)	Psych. (7)	Stroke (8)
A. Baseline Specification	-0.049** (0.019)	-0.070** (0.033)	0.024 (0.021)	0.046* (0.026)	-0.023 (0.020)	-0.051 (0.035)	-0.043 (0.032)	-0.072 (0.045)
B. Include Controls	-0.048** (0.019)	-0.075** (0.033)	0.025 (0.021)	0.039 (0.026)	-0.026 (0.019)	-0.048 (0.034)	-0.045 (0.032)	-0.067 (0.047)
C. Include Wave Fixed Effects	-0.053*** (0.019)	-0.055* (0.032)	0.008 (0.021)	0.042 (0.026)	-0.023 (0.020)	-0.050 (0.032)	-0.049 (0.031)	-0.063 (0.043)
D. Include Individual Fixed Effects	-0.045*** (0.016)	-0.082*** (0.029)	0.026 (0.018)	0.029 (0.022)	-0.028 (0.017)	-0.046* (0.027)	-0.059** (0.026)	-0.093** (0.036)
E. Use Survey Weights	0.016 (0.026)	-0.071 (0.052)	-0.022 (0.037)	0.036 (0.043)	0.004 (0.029)	-0.113* (0.061)	-0.026 (0.051)	-0.090 (0.076)
F. Survey Weights + Individual Fixed Effects	0.016 (0.019)	-0.122** (0.049)	-0.011 (0.030)	0.034 (0.037)	-0.003 (0.021)	-0.122*** (0.045)	-0.076** (0.038)	-0.048 (0.068)
G. Subsample: No Missing Values	-0.044** (0.020)	-0.068** (0.034)	0.024 (0.022)	0.063** (0.028)	-0.020 (0.021)	-0.052 (0.038)	-0.023 (0.035)	-0.116** (0.049)
H. Subsample: Balanced Panel	-0.065*** (0.024)	-0.050 (0.042)	0.024 (0.027)	0.022 (0.033)	-0.029 (0.025)	-0.031 (0.042)	-0.076* (0.040)	-0.103 (0.064)
I. Impute Proxy Interview Values	-0.054*** (0.019)	-0.078** (0.032)	0.026 (0.021)	0.045* (0.025)	-0.021 (0.020)	-0.051 (0.034)	-0.044 (0.031)	-0.087** (0.041)
J. Exclude Early 2020 Interviews	-0.045** (0.019)	-0.074** (0.034)	0.030 (0.022)	0.044 (0.027)	-0.012 (0.020)	-0.049 (0.036)	-0.024 (0.033)	-0.062 (0.049)
K. Exclude Covid Waves	-0.050** (0.022)	-0.068* (0.036)	0.015 (0.025)	0.057* (0.032)	-0.018 (0.023)	-0.059 (0.040)	-0.039 (0.038)	-0.029 (0.055)

Notes: This table reports results from assessing the robustness of our estimates for the specific health conditions and for the probability of working past 62. The estimates presented are the δ_0 s from estimating equation (2). Each column corresponds to a different health shock. Each row corresponds to a different robustness check. Row A reproduces the baseline estimates for ease of comparison. Row B adds control variables to the regressions. Row C uses wave fixed effects instead of HRS-cohort-by-wave fixed effects. Row D uses individual fixed effects and multi-wave fixed effects. Row E uses survey weights when estimating the regressions. Row F uses survey weights and includes individual fixed effects and multi-wave fixed effects. Row G focuses on a subsample of people who never have missing values for any of the outcome variables. Row H focuses on the balanced panel of observations of individuals without missing values in survey waves corresponding to event times between $\tau = -2$ and $\tau = 0$. Row I includes observations from proxy interviews by replacing the missing values with zeros. Row J excludes observations of people interviewed in March, April, or May of 2020. Row K excludes all observations in survey waves 15 and 16.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.6: Robustness of Age 65 Estimates for Specific Health Conditions

	Arthritis (1)	Cancer (2)	Diabetes (3)	Heart Disease (4)	Blood Pressure (5)	Lung Disease (6)	Psych. (7)	Stroke (8)
A. Baseline Specification	-0.024 (0.017)	-0.060** (0.030)	0.036* (0.020)	0.002 (0.025)	0.008 (0.019)	-0.045 (0.028)	-0.030 (0.030)	-0.107** (0.043)
B. Include Controls	-0.025 (0.017)	-0.063** (0.030)	0.035* (0.019)	-0.002 (0.024)	0.007 (0.019)	-0.043 (0.028)	-0.034 (0.029)	-0.108** (0.044)
C. Include Wave Fixed Effects	-0.028* (0.017)	-0.054* (0.031)	0.023 (0.019)	0.007 (0.024)	0.006 (0.018)	-0.054** (0.027)	-0.048 (0.029)	-0.095** (0.041)
D. Include Individual Fixed Effects	-0.021 (0.015)	-0.068** (0.028)	0.037** (0.017)	0.009 (0.021)	-0.000 (0.016)	-0.030 (0.025)	-0.037 (0.024)	-0.103** (0.033)
E. Use Survey Weights	0.004 (0.024)	-0.089* (0.047)	0.045 (0.036)	-0.006 (0.038)	0.035 (0.029)	-0.110** (0.051)	0.056 (0.060)	-0.100 (0.085)
F. Survey Weights + Individual Fixed Effects	0.009 (0.020)	-0.088** (0.040)	0.047 (0.030)	0.008 (0.032)	0.007 (0.022)	-0.114** (0.046)	0.001 (0.029)	-0.053 (0.045)
G. Subsample: No Missing Values	-0.020 (0.018)	-0.055* (0.030)	0.039* (0.020)	0.011 (0.026)	0.008 (0.020)	-0.051* (0.030)	-0.031 (0.031)	-0.128** (0.046)
H. Subsample: Balanced Panel	-0.021 (0.022)	-0.038 (0.035)	0.038 (0.025)	0.019 (0.030)	0.012 (0.023)	-0.036 (0.039)	-0.044 (0.037)	-0.103* (0.058)
I. Impute Proxy Interview Values	-0.028 (0.017)	-0.066** (0.029)	0.037* (0.020)	0.002 (0.024)	0.009 (0.019)	-0.044 (0.028)	-0.031 (0.029)	-0.114** (0.040)
J. Exclude Early 2020 Interviews	-0.019 (0.017)	-0.061** (0.030)	0.039* (0.020)	-0.001 (0.025)	0.014 (0.019)	-0.043 (0.028)	-0.018 (0.030)	-0.116** (0.044)
K. Exclude Covid Waves	-0.026 (0.019)	-0.059* (0.034)	0.013 (0.023)	0.021 (0.029)	-0.004 (0.021)	-0.050 (0.033)	-0.032 (0.035)	-0.111** (0.048)

Notes: This table reports results from assessing the robustness of our estimates for the specific health conditions and for the probability of working past 65. The estimates presented are the δ_0 s from estimating equation (2). Each column corresponds to a different health shock. Each row corresponds to a different robustness check. Row A reproduces the baseline estimates for ease of comparison. Row B adds control variables to the regressions. Row C uses wave fixed effects instead of HRS-cohort-by-wave fixed effects. Row D uses individual fixed effects and multi-wave fixed effects. Row E uses survey weights when estimating the regressions. Row F uses survey weights and includes individual fixed effects and multi-wave fixed effects. Row G focuses on a subsample of people who never have missing values for any of the outcome variables. Row H focuses on the balanced panel of observations of individuals without missing values in survey waves corresponding to event times between $\tau = -2$ and $\tau = 0$. Row I includes observations from proxy interviews by replacing the missing values with zeros. Row J excludes observations of people interviewed in March, April, or May of 2020. Row K excludes all observations in survey waves 15 and 16.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.7: Robustness of Age 70 Estimates for Specific Health Conditions

	Arthritis (1)	Cancer (2)	Diabetes (3)	Heart Disease (4)	Blood Pressure (5)	Lung Disease (6)	Psych. (7)	Stroke (8)
A. Baseline Specification	0.008 (0.014)	0.006 (0.024)	0.008 (0.017)	0.004 (0.021)	-0.000 (0.015)	-0.002 (0.024)	0.004 (0.024)	-0.076** (0.033)
B. Include Controls	0.006 (0.014)	0.005 (0.025)	0.004 (0.017)	0.003 (0.021)	0.001 (0.015)	0.001 (0.024)	0.004 (0.024)	-0.073** (0.035)
C. Include Wave Fixed Effects	-0.001 (0.014)	0.012 (0.024)	0.006 (0.016)	0.002 (0.020)	-0.005 (0.014)	-0.001 (0.023)	-0.018 (0.023)	-0.059* (0.034)
D. Include Individual Fixed Effects	-0.000 (0.013)	0.005 (0.024)	0.016 (0.015)	0.015 (0.019)	-0.002 (0.013)	0.001 (0.020)	-0.002 (0.021)	-0.058* (0.031)
E. Use Survey Weights	0.040** (0.020)	0.002 (0.032)	0.017 (0.026)	0.013 (0.030)	0.000 (0.023)	-0.014 (0.034)	0.063 (0.045)	-0.049 (0.055)
F. Survey Weights + Individual Fixed Effects	0.027 (0.017)	0.002 (0.028)	0.021 (0.021)	0.025 (0.027)	-0.011 (0.018)	-0.010 (0.028)	0.009 (0.028)	0.004 (0.028)
G. Subsample: No Missing Values	0.014 (0.014)	0.001 (0.025)	0.012 (0.019)	0.009 (0.022)	-0.007 (0.015)	-0.000 (0.028)	-0.007 (0.025)	-0.082** (0.037)
H. Subsample: Balanced Panel	-0.007 (0.015)	0.044 (0.036)	0.005 (0.022)	0.002 (0.024)	0.001 (0.019)	0.001 (0.030)	0.013 (0.028)	-0.055 (0.057)
I. Impute Proxy Interview Values	0.006 (0.014)	0.003 (0.024)	0.009 (0.017)	0.004 (0.020)	0.001 (0.015)	-0.001 (0.024)	0.004 (0.024)	-0.075** (0.032)
J. Exclude Early 2020 Interviews	0.009 (0.014)	0.012 (0.025)	0.010 (0.017)	0.003 (0.021)	0.003 (0.015)	-0.003 (0.024)	0.012 (0.024)	-0.082** (0.035)
K. Exclude Covid Waves	0.007 (0.015)	0.015 (0.029)	-0.013 (0.019)	0.007 (0.025)	-0.010 (0.017)	-0.012 (0.024)	0.015 (0.026)	-0.057 (0.043)

Notes: This table reports results from assessing the robustness of our estimates for the specific health conditions and for the probability of working past 70. The estimates presented are the δ_0 s from estimating equation (2). Each column corresponds to a different health shock. Each row corresponds to a different robustness check. Row A reproduces the baseline estimates for ease of comparison. Row B adds control variables to the regressions. Row C uses wave fixed effects instead of HRS-cohort-by-wave fixed effects. Row D uses individual fixed effects and multi-wave fixed effects. Row E uses survey weights when estimating the regressions. Row F uses survey weights and includes individual fixed effects and multi-wave fixed effects. Row G focuses on a subsample of people who never have missing values for any of the outcome variables. Row H focuses on the balanced panel of observations of individuals without missing values in survey waves corresponding to event times between $\tau = -2$ and $\tau = 0$. Row I includes observations from proxy interviews by replacing the missing values with zeros. Row J excludes observations of people interviewed in March, April, or May of 2020. Row K excludes all observations in survey waves 15 and 16.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$